

PROJECT DOCUMENTATION

Nexus Service Marketing System

A web application for telephone and internet services — Dial-Up, Broadband and Landline — covering retail shops, orders, feasibility checks, connections, equipment stock, postpaid billing and payments.

ASP.NET Core 8 MVC

C#

Entity Framework Core 8

SQL Server

Bootstrap 5

HTML5 · CSS3 · JavaScript

SUBMITTED BY

Student name _____

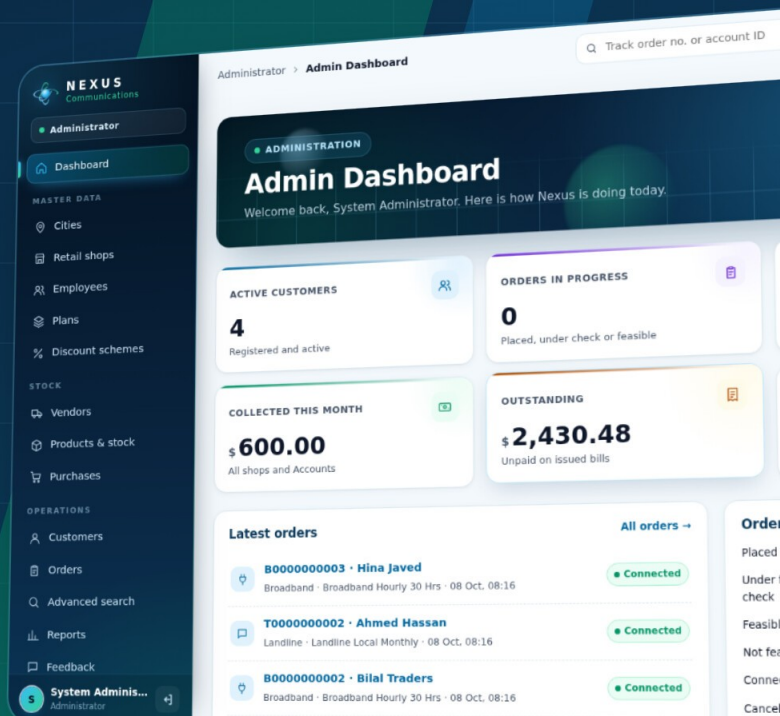
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I am also thankful to my family and friends for their encouragement throughout the project.

Certificate

This is to certify that the eProject titled "**Nexus Service Marketing System**" has been completed by the student named on the cover page, as part of the curriculum, under the guidance of the faculty named there.

Faculty signature

Centre manager

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1. Introduction

Nexus Communications provides telephone and internet services - Dial-Up, Broadband and Landline - through retail shops in several cities. The **Nexus Service Marketing System** is a web application that runs the whole business in one place: plans and prices, retail shops, employees, customers, orders, feasibility checks, connections, equipment stock, postpaid bills and payments.

Every user signs in to a dashboard for their own role. Customers can register, order and follow their order online; retail staff serve walk-in customers and collect payments; the technical team checks feasibility and creates connections; accounts staff generate bills; and the administrator maintains the master data and watches the whole operation.

1.1 Objectives

- Replace manual registers with one database that every department shares.
- Give each role exactly the screens it needs, protected by role-based login.
- Generate order numbers and account IDs automatically in the format of the specification.
- Calculate bills correctly: plan fee, call charges, usage, deposit, bulk discount and 12.24 % service tax.
- Show the status of every order, connection and bill at any time, including amounts due.
- Present it in a modern, fast and easy interface that also works on tablets and phones.

1.2 Scope

Table 1. *Project scope*

In scope	Out of scope
Dial-Up, Broadband and Landline orders and connections	Online card payment gateway (payments are recorded by staff)
Five user roles + public visitors	Physical network / exchange equipment management
Feasibility checks, provisioning, equipment issue and replacement	SMS / e-mail notifications
Postpaid billing, partial payments, overdue suspension	Payroll and HR
Advanced search, tracking, reports, customer documents, feedback	Mobile app (the website is responsive instead)

1.3 Technology stack

Table 2. *Technology stack*

Layer	Technology
Front end	Razor views, HTML5, CSS3, JavaScript, Bootstrap 5, Google Fonts (Space Grotesk, DM Sans)
Back end	ASP.NET Core 8 MVC (C# 12), Areas per role, cookie authentication, role authorization
Data access	Entity Framework Core 8 (code-first, 5 migrations), LINQ
Database	Microsoft SQL Server 2019 / 2022 (any edition, incl. Express)
Tools	Visual Studio 2022, SQL Server Management Studio, .NET 8 SDK, dotnet-ef, Google Chrome

1.4 Project plan

Table 3. *Project plan*

Phase	Work	Output
1. Analysis	Study the problem statement, roles and business rules	Requirement list
2. Design	ER diagram, tables, ID formats, screens, interface design	Database design, GUI standards
3. Database	EF Core entities, configurations, migrations, seed data	19 tables in SQL Server
4. Development	Five role areas; services for orders, connections and billing	Working website
5. UI design	App shell, dashboards, motion, overlays, responsive layout	site.css / nexus-app.css, site.js / nexus-app.js
6. Testing	Automated browser workflow, negative tests, unit checklist	Test results (chapter 8)
7. Delivery	Documentation, presentation, database backup, source ZIP	Final submission

2. Problem Definition

Nexus Communications sells Dial-Up, Broadband and Landline connections from retail shops in several cities. Today each shop keeps its own registers for customers and orders, the technical team receives orders on paper, and the accounts department prepares bills by hand. This causes:

- **Lost and duplicated orders** - no single order number that every department can follow.
- **Slow feasibility and installation** - technical staff do not see new orders until paperwork arrives.
- **Billing mistakes** - plan fees, call charges, deposits, bulk discounts and the 12.24 % service tax are calculated by hand.
- **No view of dues** - shops cannot see what a customer owes, and overdue connections stay active.
- **Equipment losses** - modems and routers issued to customers are not tracked against stock.
- **No self-service** - customers must visit or phone a shop to know the status of their order.

The company needs a central, role-based web system where every order gets a unique number, moves through feasibility, connection, billing and payment in one database, and where every user sees the information they need instantly.

Identifier formats required by the specification

Order number - 11 characters: service letter (D Dial-Up, B Broadband, T Telephone) + 10 digits, e.g. B00000000001.

Account ID - 16 characters: service letter + 3-digit city code + 12-digit serial, e.g. B00100000000000001.

Both are generated by the server and protected by CHECK constraints in the database.

3. Customer Requirements Specification

3.1 Users and roles

Table 4. *Users and roles*

Role	Who	Main responsibilities
Visitor	Anyone	View plans and prices, track an order / account, register as a customer
Customer	Registered customer	Profile, place orders, follow orders, connections, bills and amount due, documents, feedback
Retail staff	Shop employee	Register walk-in customers, place orders, view shop connections, collect payments, payment history, search
Technical	Technical team	Feasibility checks, create connections, issue / replace equipment, change plan, suspend overdue and reactivate, stock
Accounts	Accounts department	Generate bills, record payments, view unpaid and overdue bills, search
Administrator	Head office	Cities, shops, employees and logins, plans, discount schemes, vendors, products, purchases, customers, orders, reports, feedback

3.2 Functional requirements

Table 5. *Functional requirements*

ID	Requirement	Role
FR-01	Secure login for every role; passwords stored as salted hashes; change password	All
FR-02	Maintain cities (unique 3-digit code) and retail shops	Admin
FR-03	Maintain employees with a role, shop (for retail staff) and login	Admin
FR-04	Add, edit, search and delete plans for Dial-Up, Broadband and Landline	Admin
FR-05	Maintain bulk discount bands (10-14: 25 %, 15-24: 50 %, 25-49: 75 %, 50+: 100 %)	Admin
FR-06	Maintain vendors, products and stock; record purchases that increase stock	Admin
FR-07	Register customers (individual or corporate) with a login	Retail, Customer, Admin
FR-08	Place an order; generate the 11-character order number; apply bulk discount	Retail, Customer
FR-09	Dial-up needs a Nexus landline: order both together or link an existing line	Retail, Customer, Technical
FR-10	Record landline and internet feasibility (distance, server availability, result)	Technical
FR-11	Create connections from feasible orders; generate account IDs; issue equipment from stock	Technical
FR-12	Set connections temporarily or permanently inactive; change plan; replace spoiled equipment	Technical
FR-13	Generate bills: plan fee per validity period, call charges, usage, deposit, discount, 12.24 % tax, brought-forward balance	Accounts

ID	Requirement	Role
FR-14	Record full or partial payments; never more than the outstanding amount	Accounts, Retail
FR-15	List active connections with overdue bills; suspend them; reactivate when paid	Technical
FR-16	Track an order number or account ID publicly; amounts only after sign-in	All
FR-17	Advanced search by ID, name, contact, connection type and date range	Admin, Retail, Technical, Accounts
FR-18	Upload private customer documents (PDF / PNG / JPEG, 5 MB) filed by year and city	Customer, Retail, Admin
FR-19	Collect customer feedback; reports for the administrator	Customer, Admin
FR-20	Role dashboards with live figures and items that need attention	All signed-in users

3.3 Non-functional requirements

Table 6. *Non-functional requirements*

Area	Requirement
Security	Cookie authentication, role-based authorization on every controller, anti-forgery tokens on every form, PBKDF2 password hashes, private document storage, customers can only open their own records
Data integrity	Primary / foreign keys, 47 CHECK constraints, unique indexes, row-version concurrency on orders and connections, no cascade deletes
Usability	Sidebar navigation per role, dashboards, consistent status colours, inline validation, confirm dialogs before destructive actions
Performance	Server-side paging limits (300 rows), AsNoTracking reads, indexes on foreign keys and identifiers
Portability	Runs on any SQL Server edition; one connection string in appsettings.json; database created by migrations or one SQL script
Accessibility	Keyboard focus styles, skip link, ARIA labels, WCAG contrast for text, animations disabled for "reduced motion" users
Responsiveness	Works from 360 px phones to wide desktop screens

3.4 Business rules

- A plan fee is charged once per plan validity period (e.g. a quarterly plan every 3 months).
- Call charges = local, STD and messaging-for-mobile minutes x the plan's per-minute rates.
- The security deposit and the bulk discount appear on the first bill only.
- Service tax of **12.24 %** is applied to the whole bill.
- The unpaid amount of earlier bills is shown as **brought forward**.
- A bill past its due date and not fully paid makes an active connection **overdue**; Technical can set it temporarily inactive, and reactivate it once all bills are paid.
- Equipment replaced because the customer spoiled it is charged on the next bill (once).
- A multi-connection order records device serials after provisioning, one per connection.

3.5 Requirement traceability

Table 7. Requirement traceability

No.	Requirement (from the problem statement)	Where implemented
1	Separate logins for Admin, Accounts, Technical, Retail employees and Customers	Role-based Areas, cookie login, hashed passwords
2	Admin maintains employees, stock, vendors, retail shops, plans (insert, update, delete, search)	Admin area
3	Order ID: 11 characters, first letter D / T / B + serial (e.g. D0000000001)	IdentifierGenerator, DB check constraint
4	Account ID: type letter + 3-digit city code + 12-digit serial (16 characters)	IdentifierGenerator, Cities.Code
5	Only retail employees (and customers themselves) place orders	Retail and Customer areas
6	Feasibility: landline + internet for Dial-Up; internet only if the customer already has a Nexus landline	OrderWorkflowService
7	Technical updates orders, creates connections, sets them temporarily / permanently inactive, maintains equipment	Technical area, ConnectionProvisioningService
8	Accounts generate bills; payments recorded by Accounts and Retail staff	Accounts and Retail areas, BillingService
9	Bulk discount 10-14: 25 %, 15-24: 50 %, 25-49: 75 %, 50+: 100 % on advance and deposit	DiscountSchemes, OrderWorkflowService
10	Service tax 12.24 % on the whole bill	BillingService
11	Search order status by order ID and connection status (with due amount) by account ID	Public Track page, quick Track box
12	Advanced search by ID, name, connection type, date / period, contact number	Admin and staff Advanced search
13	Customer sees contact details, places orders, tracks orders, views bills	Customer area and dashboard
14	Address proofs, application forms and receipts filed by year and area	Customer Documents
15	Customer feedback; customers can change plan	Feedback page; Technical > Change plan
16	Plan details with security deposits and call charges	Plans table (18 plans seeded), public Plans page

3.6 Plan price list (seeded)

Table 8. *Plan price list*

Type	Plan	Price	Security deposit
Dial-Up	Hourly 10 / 30 / 60 hrs	\$50 / \$130 / \$260	\$325
Dial-Up	Unlimited 28 Kbps monthly / quarterly	\$75 / \$150	\$325
Dial-Up	Unlimited 56 Kbps monthly / quarterly	\$100 / \$180	\$325
Broadband	Hourly 30 / 60 hrs	\$175 / \$315	\$500
Broadband	Unlimited 64 Kbps monthly / quarterly	\$225 / \$400	\$500
Broadband	Unlimited 128 Kbps monthly / quarterly	\$350 / \$445	\$500
Landline	Local: yearly rental (\$0.55 / min), monthly (\$0.75 / min)	\$75 / \$35	\$250
Landline	STD: monthly, half-yearly, yearly (yearly price not given in the specification; seeded inactive)	\$125 / \$420 / -	\$250

4. System Design

4.1 Architecture

The application follows a **three-layer architecture**. The presentation layer (areas, controllers and Razor views) never talks to the database directly; it calls services, which hold every business rule and use Entity Framework Core to read and write SQL Server.

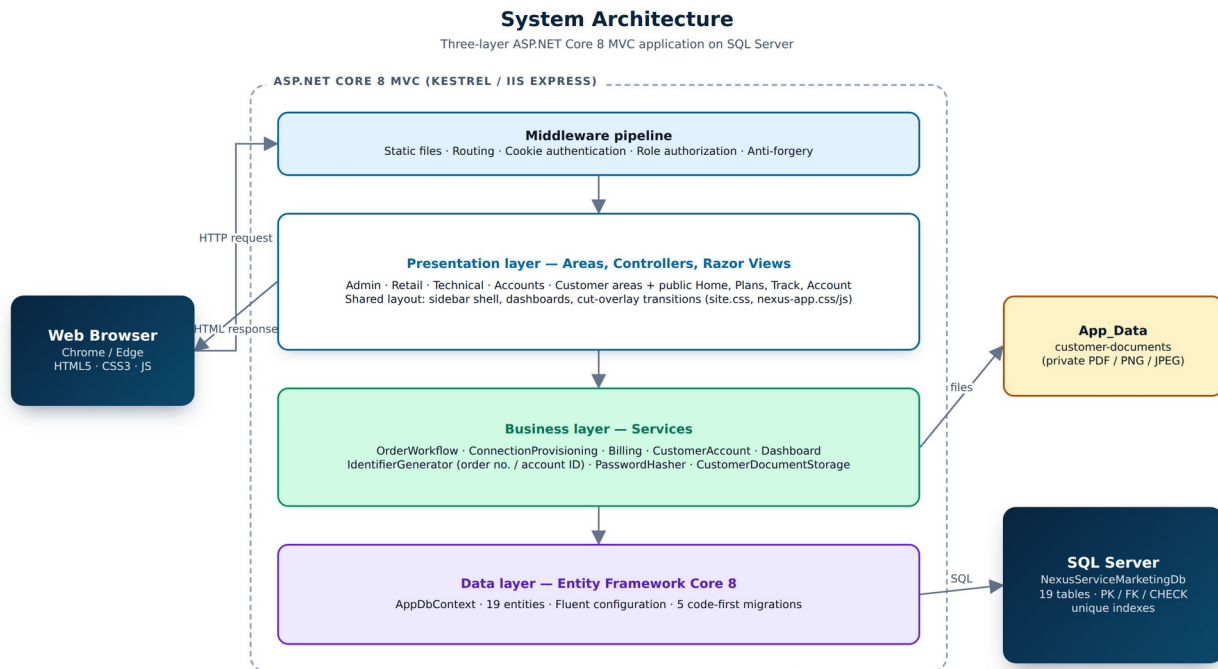


Figure 1. System architecture

4.2 Modules

Table 9. Application modules

Module (Area)	Controllers	Purpose
Public	Home, Catalog, Track, Account	Landing page, plans and prices, order / account tracking, login, registration, change password
Admin	Home, Cities, RetailShops, Employees, Plans, DiscountSchemes, Vendors, Products, Purchases, Customers, Operations	Master data, stock, customers, orders, advanced search, reports, feedback
Retail	Home, Customers, Orders, Connections, Payments	Walk-in customers and orders, shop connections, collect payments, payment history
Technical	Home, Feasibility, Connections, Products	Feasibility queue, provisioning, status changes, overdue list, equipment and stock
Accounts	Home, Bills	Bill generation with charge preview, payments
Customer	Home, Orders, Portal, Documents	Self-service orders, connections, bills, profile, documents, feedback
Search	Search	Advanced search shared by Retail (own shop), Technical and Accounts

4.3 Services (business layer)

Table 10. Business services

Service	Responsibility
OrderWorkflowService	Validates and places orders, applies discount bands, dial-up + landline rule, starts and records feasibility checks, cancels orders
ConnectionProvisioningService	Creates connections and account IDs, issues equipment, changes status / plan, replaces equipment, finds overdue and cleared connections
BillingService	Previews and generates bills (plan fee, calls, usage, deposit, discount, 12.24 % tax, brought-forward), records payments
CustomerAccountService	Registers customers with a login, updates profiles
IdentifierGenerator	Thread-safe order numbers and account IDs using the IdentifierCounters table
DashboardService	Read-only figures for the five role dashboards
PasswordHasherService / CustomerDocumentStorage	Password hashing; private file storage under App_Data

4.4 Security design

- Each area controller carries `[Authorize(Roles = ...)]`; anonymous users are redirected to Login and wrong roles to Access denied.
- Every POST form has an anti-forgery token (`[ValidateAntiForgeryToken]`).
- Passwords are hashed with PBKDF2 and a random salt; the admin account is created at first start and should change its password.

- Customers, retail shops and employees see only their own data (shop and customer filters in every query).
- Customer documents are stored outside wwwroot and are downloaded only through an authorized action.

5. Database Design

The database **NexusServiceMarketingDb** has 19 tables. It is created by five Entity Framework migrations ([InitialCreate](#), [CustomerSelfServiceOrders](#), [VendorProductPurchases](#), [CustomerDocumentFiling](#), [SpecGapFixes](#)) or by the equivalent script [Database/NexusServiceMarketingDb-schema.sql](#). The migrations also seed 18 tariff plans and 4 bulk-discount bands.

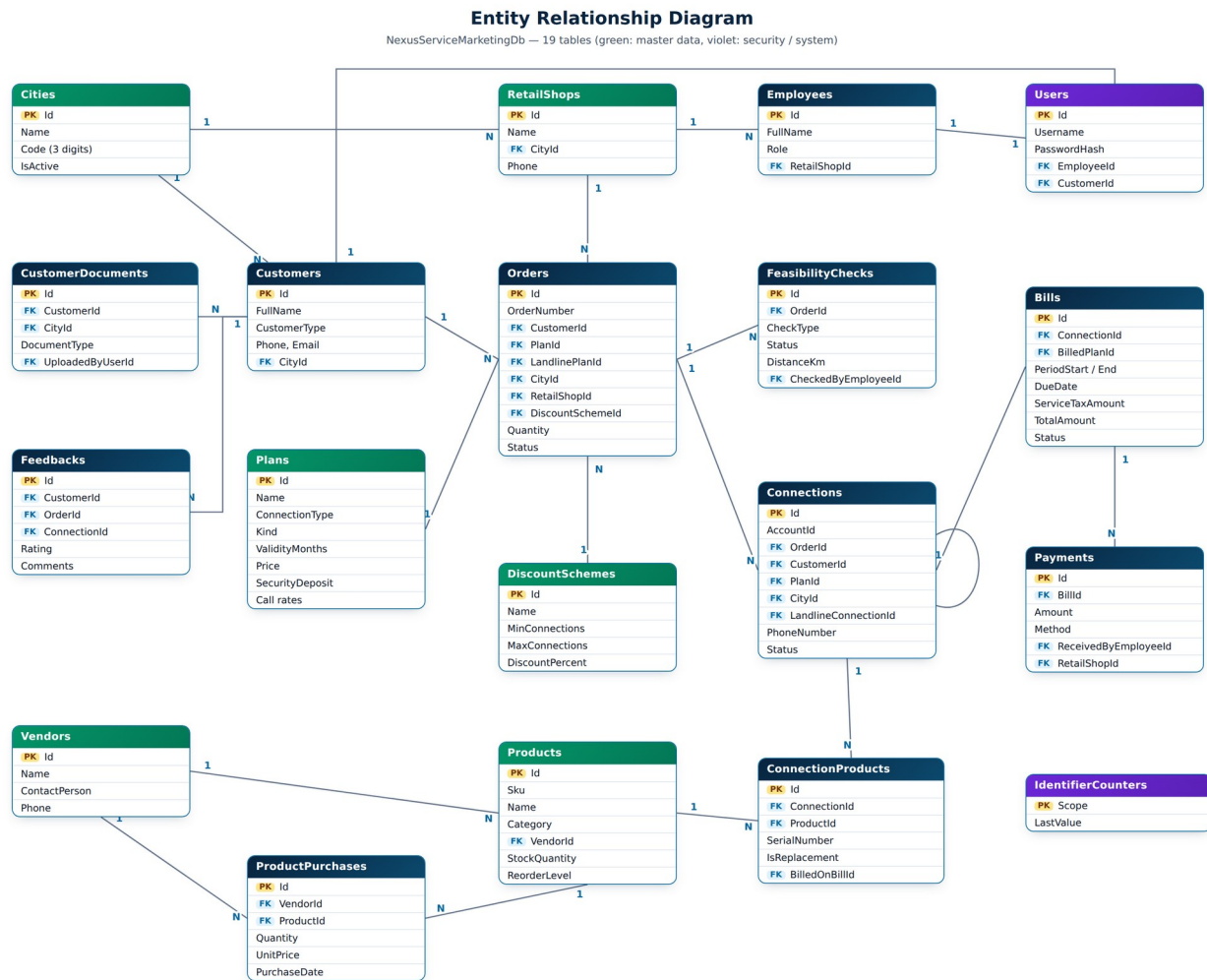


Figure 2. Entity relationship diagram

5.1 Tables

Table 11. Table groups

Group	Tables
Master data	Cities, RetailShops, Plans, DiscountSchemes, Vendors, Products
People and security	Employees, Customers, Users
Orders and connections	Orders, FeasibilityChecks, Connections, ConnectionProducts
Billing	Bills, Payments
Stock, documents, other	ProductPurchases, CustomerDocuments, Feedbacks, IdentifierCounters

5.2 Integrity rules

- Every relationship is a foreign key with **ON DELETE NO ACTION**; records in use are deactivated (`IsActive = 0`) instead of deleted.
- Unique indexes: OrderNumber, AccountId, Username, city Code and Name, product Sku, plan Name, vendor Name, one bill per connection per period.
- CHECK constraints guard identifier formats, status values, amounts (≥ 0), percentages (0-100) and plan kinds.
- Money uses `decimal(18,2)`; call rates `decimal(10,4)`; dates `date` / `datetime2` in UTC.

5.3 Data dictionary

The dictionary below was exported from the live database, so it matches the delivered schema exactly.

Bills

Postpaid bill of one connection for one billing period, with every charge stored as issued.

Column	Data type	Null	Key	Description
Id	int	No	PK	Primary key (identity)
ConnectionId	int	No	FK	References Connections.Id
PeriodStart	date	No		
PeriodEnd	date	No		
IssueDate	date	No		
DueDate	date	No		Pay-by date; unpaid after this = overdue
PlanCharge	decimal(18,2)	No		
UsageCharge	decimal(18,2)	No		
SecurityDepositCharge	decimal(18,2)	No		
ReplacementCharge	decimal(18,2)	No		
DiscountAmount	decimal(18,2)	No		
SubTotal	decimal(18,2)	No		
ServiceTaxRate	decimal(5,2)	No		Service tax rate (12.24 %)
ServiceTaxAmount	decimal(18,2)	No		Tax amount on the bill

Column	Data type	Null	Key	Description
TotalAmount	decimal(18,2)	No		Bill total after tax
Status	varchar(30)	No		Workflow status (CHECK list of values)
GeneratedByEmployeeId	int	No	FK	References Employees.Id
GeneratedAtUtc	datetime2	No		
RowVersion	rowversion	No		Concurrency token (row version)
BilledPlanId	int	Yes	FK	References Plans.Id
CallCharge	decimal(18,2)	No		
LocalMinutes	int	No		
MobileMinutes	int	No		
PlanValidUntil	date	Yes		
PreviousBalance	decimal(18,2)	No		Unpaid amount brought forward from earlier bills
StdMinutes	int	No		

Cities

Cities served; the 3-digit code is part of every account ID.

Column	Data type	Null	Key	Description
Id	int	No	PK	Primary key (identity)
Name	nvarchar(100)	No		Display name
Code	char(3)	No		3-digit city code (CHECK: [0-9][0-9][0-9])
IsActive	bit	No		Inactive records are hidden from new work (soft delete)

ConnectionProducts

Equipment (modem, router, handset) issued to a connection, including replacements.

Column	Data type	Null	Key	Description
Id	int	No	PK	Primary key (identity)
ConnectionId	int	No	FK	References Connections.Id
ProductId	int	No	FK	References Products.Id
Quantity	int	No		Number of connections / items (≥ 1)
SerialNumber	nvarchar(60)	Yes		Device serial number (unique per product)
IsReplacement	bit	No		
IssuedAtUtc	datetime2	No		
ReturnedAtUtc	datetime2	Yes		
Notes	nvarchar(500)	Yes		
BilledOnBillId	int	Yes	FK	References Bills.Id

Column	Data type	Null	Key	Description
ReplacementChargeAmount	decimal(18,2)	No		

Connections

A working Dial-Up, Broadband or Landline connection created from a feasible order.

Column	Data type	Null	Key	Description
Id	int	No	PK	Primary key (identity)
AccountId	char(16)	No		D/B/T + 3-digit city + 12-digit serial, unique (CHECK)
OrderId	int	No	FK	References Orders.Id
CustomerId	int	No	FK	References Customers.Id
PlanId	int	No	FK	References Plans.Id
ConnectionType	char(1)	No		Service letter: D = Dial-Up, B = Broadband, T = Telephone (CHECK)
CityId	int	No	FK	References Cities.Id
InstallationAddress	nvarchar(300)	No		Where the connection is installed
PhoneNumber	nvarchar(20)	Yes		Telephone number of a landline
LandlineConnectionId	int	Yes	FK	References Connections.Id
Status	varchar(30)	No		Workflow status (CHECK list of values)
SecurityDepositAmount	decimal(18,2)	No		
CreatedByEmployeeId	int	No	FK	References Employees.Id
ActivatedAtUtc	datetime2	No		
StatusChangedAtUtc	datetime2	No		
RowVersion	rowversion	No		Concurrency token (row version)

CustomerDocuments

Private customer files (address proof, application form, receipts) filed by year and city.

Column	Data type	Null	Key	Description
Id	int	No	PK	Primary key (identity)
CustomerId	int	No	FK	References Customers.Id
CityId	int	No	FK	References Cities.Id
DocumentType	varchar(30)	No		
DocumentYear	int	No		
OriginalFileName	nvarchar(255)	No		
StorageName	nvarchar(80)	No		Random file name on disk (private)
ContentType	nvarchar(100)	No		

Column	Data type	Null	Key	Description
SizeBytes	bigint	No		File size (max 5 MB)
Notes	nvarchar(500)	Yes		
UploadedByUserId	int	No	FK	References Users.Id
UploadedAtUtc	datetime2	No		

Customers

Individual or corporate customers.

Column	Data type	Null	Key	Description
Id	int	No	PK	Primary key (identity)
FullName	nvarchar(150)	No		Person or company name
CustomerType	varchar(30)	No		
CompanyName	nvarchar(150)	Yes		
Email	nvarchar(256)	Yes		E-mail address
Phone	nvarchar(20)	No		Contact phone
AddressLine	nvarchar(300)	No		
PostalCode	nvarchar(20)	Yes		
CityId	int	No	FK	References Cities.Id
IsActive	bit	No		Inactive records are hidden from new work (soft delete)
CreatedAtUtc	datetime2	No		Created at (UTC)

DiscountSchemes

Bulk / corporate discount bands by number of connections.

Column	Data type	Null	Key	Description
Id	int	No	PK	Primary key (identity)
Name	nvarchar(100)	No		Display name
MinConnections	int	No		
MaxConnections	int	Yes		
DiscountPercent	decimal(5,2)	No		Percentage 0-100 (CHECK)
AppliesToAdvance	bit	No		
AppliesToSecurityDeposit	bit	No		
IsActive	bit	No		Inactive records are hidden from new work (soft delete)
CreatedAtUtc	datetime2	No		Created at (UTC)

Employees

Staff of Nexus: Admin, Accounts, Technical and Retail staff.

Column	Data type	Null	Key	Description
Id	int	No	PK	Primary key (identity)
FullName	nvarchar(150)	No		Person or company name
Email	nvarchar(256)	No		E-mail address
Phone	nvarchar(20)	No		Contact phone
Role	varchar(30)	No		Admin, Accounts, Technical or RetailStaff
RetailShopId	int	Yes	FK	References RetailShops.Id
IsActive	bit	No		Inactive records are hidden from new work (soft delete)
CreatedAtUtc	datetime2	No		Created at (UTC)

FeasibilityChecks

Landline and / or internet feasibility check of an order.

Column	Data type	Null	Key	Description
Id	int	No	PK	Primary key (identity)
OrderId	int	No	FK	References Orders.Id
CheckType	varchar(30)	No		Landline or Internet
Status	varchar(30)	No		Workflow status (CHECK list of values)
DistanceKm	decimal(8,2)	Yes		Distance from exchange / server (km)
ServerAvailable	bit	Yes		Internet server port available
Remarks	nvarchar(500)	Yes		
CheckedByEmployeeId	int	Yes	FK	References Employees.Id
CheckedAtUtc	datetime2	Yes		
CreatedAtUtc	datetime2	No		Created at (UTC)

Feedbacks

Customer feedback with a 1-5 rating.

Column	Data type	Null	Key	Description
Id	int	No	PK	Primary key (identity)
CustomerId	int	No	FK	References Customers.Id
OrderId	int	Yes	FK	References Orders.Id
ConnectionId	int	Yes	FK	References Connections.Id
Rating	int	No		Rating 1-5
Comments	nvarchar(2000)	No		
SubmittedAtUtc	datetime2	No		

IdentifierCounters

Running counters used to generate order numbers and account IDs safely.

Column	Data type	Null	Key	Description
Scope	varchar(40)	No	PK	Counter name (e.g. order type)
LastValue	bigint	No		Last number issued

Orders

A request for one or more connections of one service type.

Column	Data type	Null	Key	Description
Id	int	No	PK	Primary key (identity)
OrderNumber	char(11)	No		D/B/T + 10 digits, unique (CHECK)
CustomerId	int	No	FK	References Customers.Id
ConnectionType	char(1)	No		Service letter: D = Dial-Up, B = Broadband, T = Telephone (CHECK)
PlanId	int	No	FK	References Plans.Id
CityId	int	No	FK	References Cities.Id
InstallationAddress	nvarchar(300)	No		Where the connection is installed
Quantity	int	No		Number of connections / items (>= 1)
RetailShopId	int	Yes	FK	References RetailShops.Id
PlacedByEmployeeId	int	Yes	FK	References Employees.Id
DiscountSchemeId	int	Yes	FK	References DiscountSchemes.Id
DiscountPercent	decimal(5,2)	No		Percentage 0-100 (CHECK)
Status	varchar(30)	No		Workflow status (CHECK list of values)
PlacedAtUtc	datetime2	No		
StatusChangedAtUtc	datetime2	No		

Column	Data type	Null	Key	Description
Remarks	nvarchar(500)	Yes		
RowVersion	rowversion	No		Concurrency token (row version)
LandlinePlanId	int	Yes	FK	References Plans.Id

Payments

Full or partial payment against a bill, at a retail shop or at Accounts.

Column	Data type	Null	Key	Description
Id	int	No	PK	Primary key (identity)
BillId	int	No	FK	References Bills.Id
Amount	decimal(18,2)	No		Amount paid
Method	varchar(30)	No		Cash, Card, BankTransfer, Cheque or Online
Reference	nvarchar(100)	Yes		
PaidAtUtc	datetime2	No		
ReceivedByEmployeeId	int	No	FK	References Employees.Id
RetailShopId	int	Yes	FK	References RetailShops.Id

Plans

Tariff plans for Dial-Up, Broadband and Landline with prices, deposits and call rates.

Column	Data type	Null	Key	Description
Id	int	No	PK	Primary key (identity)
Name	nvarchar(150)	No		Display name
Description	nvarchar(500)	Yes		
ConnectionType	char(1)	No		Service letter: D = Dial-Up, B = Broadband, T = Telephone (CHECK)
Kind	varchar(30)	No		Hourly, Unlimited, LocalRental or StdRental (CHECK)
IncludedHours	int	Yes		
SpeedKbps	int	Yes		
ValidityMonths	int	No		Plan validity (> 0)
Price	decimal(18,2)	No		Plan fee per validity period
SecurityDeposit	decimal(18,2)	No		Refundable deposit charged on the first bill
LocalCallRatePerMinute	decimal(10,4)	Yes		
StdCallRatePerMinute	decimal(10,4)	Yes		
MobileMessagingRatePerMinute	decimal(10,4)	Yes		

Column	Data type	Null	Key	Description
IsActive	bit	No		Inactive records are hidden from new work (soft delete)
CreatedAtUtc	datetime2	No		Created at (UTC)

ProductPurchases

Equipment bought from a vendor; receipts increase stock.

Column	Data type	Null	Key	Description
Id	int	No	PK	Primary key (identity)
VendorId	int	No	FK	References Vendors.Id
ProductId	int	No	FK	References Products.Id
Quantity	int	No		Number of connections / items (≥ 1)
UnitPrice	decimal(18,2)	No		
AmountPaid	decimal(18,2)	No		
PurchaseDate	date	No		
SupplierReference	nvarchar(100)	Yes		
Notes	nvarchar(500)	Yes		
RecordedByEmployeeId	int	No	FK	References Employees.Id
RecordedAtUtc	datetime2	No		

Products

Equipment catalogue and stock level.

Column	Data type	Null	Key	Description
Id	int	No	PK	Primary key (identity)
Sku	nvarchar(40)	No		Unique stock-keeping unit
Name	nvarchar(150)	No		Display name
Category	varchar(30)	No		
Description	nvarchar(500)	Yes		
VendorId	int	No	FK	References Vendors.Id
PurchasePrice	decimal(18,2)	No		
ReplacementCharge	decimal(18,2)	No		
StockQuantity	int	No		Units in stock
ReorderLevel	int	No		Low-stock warning level
IsActive	bit	No		Inactive records are hidden from new work (soft delete)
CreatedAtUtc	datetime2	No		Created at (UTC)

Column	Data type	Null	Key	Description
RowVersion	rowversion	No		Concurrency token (row version)

RetailShops

Nexus retail outlets, one or more per city.

Column	Data type	Null	Key	Description
Id	int	No	PK	Primary key (identity)
Name	nvarchar(150)	No		Display name
AddressLine	nvarchar(300)	No		
Phone	nvarchar(20)	Yes		Contact phone
CityId	int	No	FK	References Cities.Id
IsActive	bit	No		Inactive records are hidden from new work (soft delete)

Users

Login accounts (username + hashed password) linked to an employee or a customer.

Column	Data type	Null	Key	Description
Id	int	No	PK	Primary key (identity)
Username	nvarchar(50)	No		Unique login name
PasswordHash	nvarchar(500)	No		PBKDF2 password hash - never the plain password
IsActive	bit	No		Inactive records are hidden from new work (soft delete)
CreatedAtUtc	datetime2	No		Created at (UTC)
LastLoginAtUtc	datetime2	Yes		
EmployeeId	int	Yes	FK	References Employees.Id
CustomerId	int	Yes	FK	References Customers.Id

Vendors

Equipment manufacturers and suppliers.

Column	Data type	Null	Key	Description
Id	int	No	PK	Primary key (identity)
Name	nvarchar(150)	No		Display name
ContactPerson	nvarchar(100)	Yes		
Email	nvarchar(256)	Yes		E-mail address
Phone	nvarchar(20)	No		Contact phone
AddressLine	nvarchar(300)	Yes		
IsActive	bit	No		Inactive records are hidden from new work (soft delete)
CreatedAtUtc	datetime2	No		Created at (UTC)

6. Process Design

6.1 Context diagram (DFD level 0)

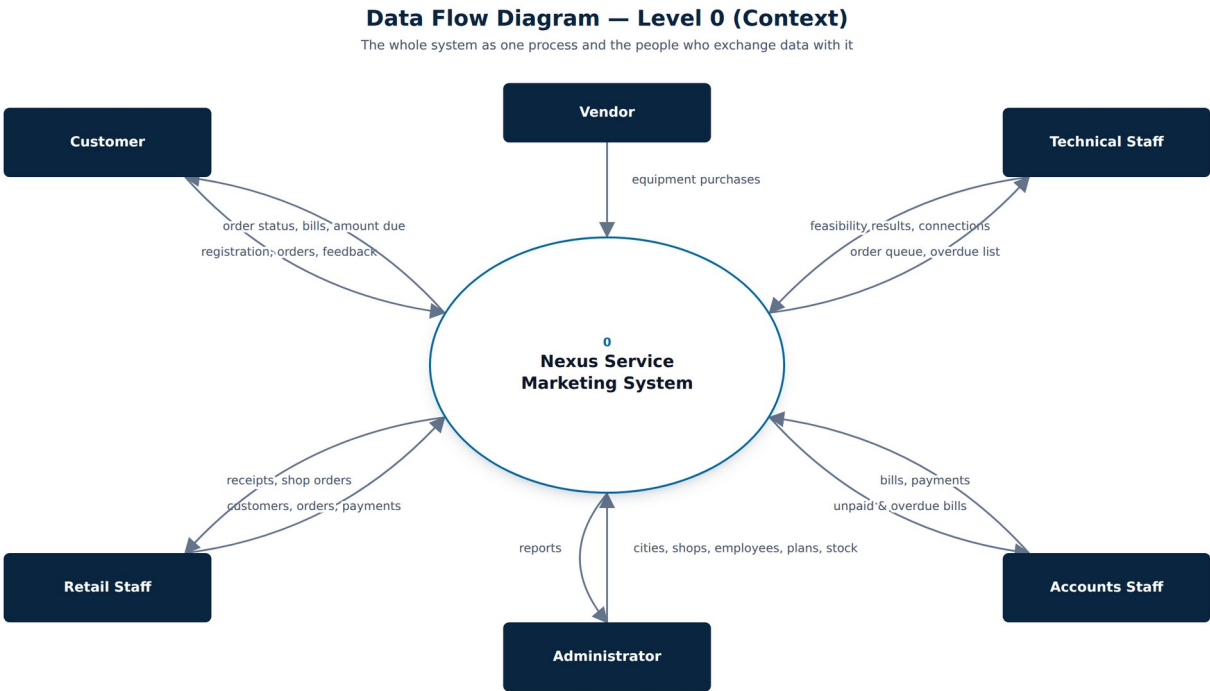


Figure 3. Data flow diagram - level 0

6.2 DFD level 1

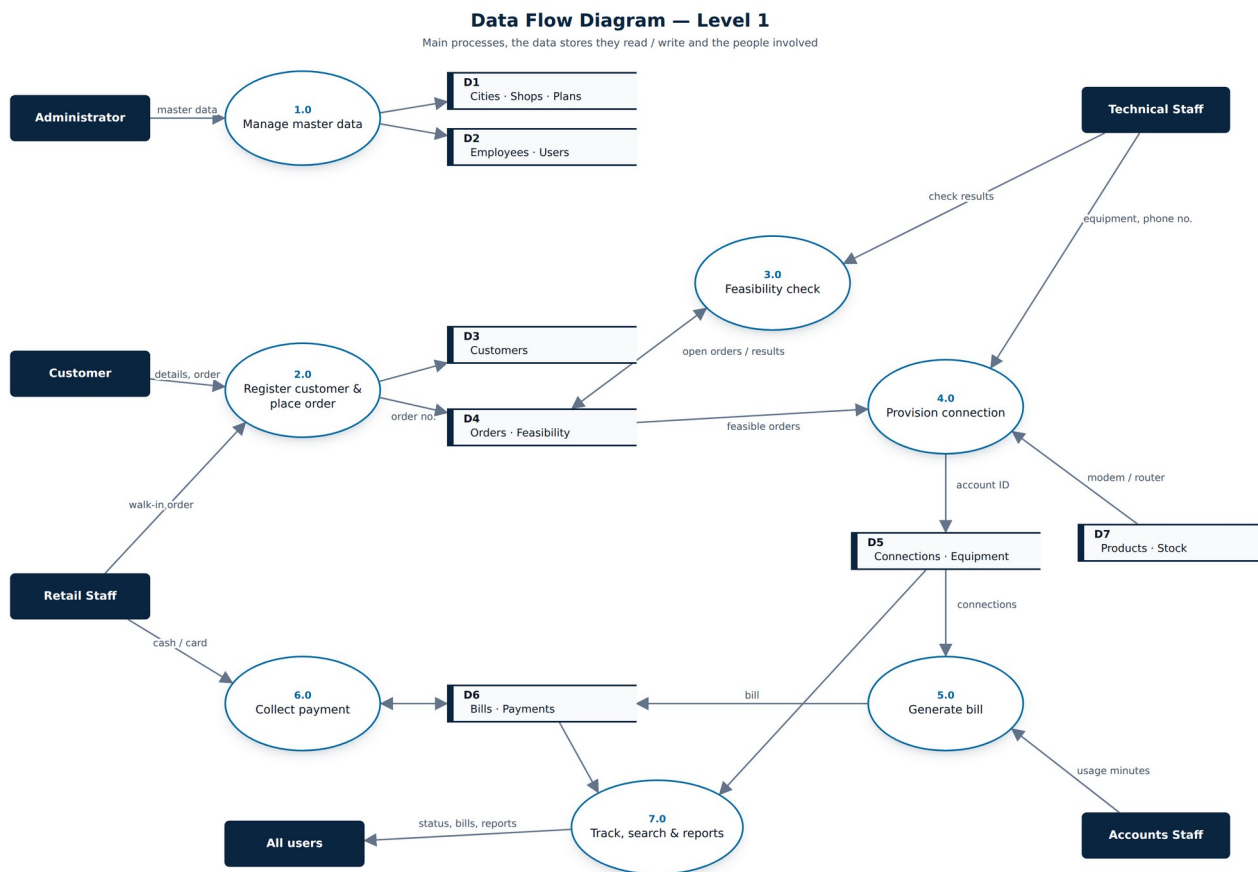


Figure 4. Data flow diagram - level 1

6.3 Use case diagram

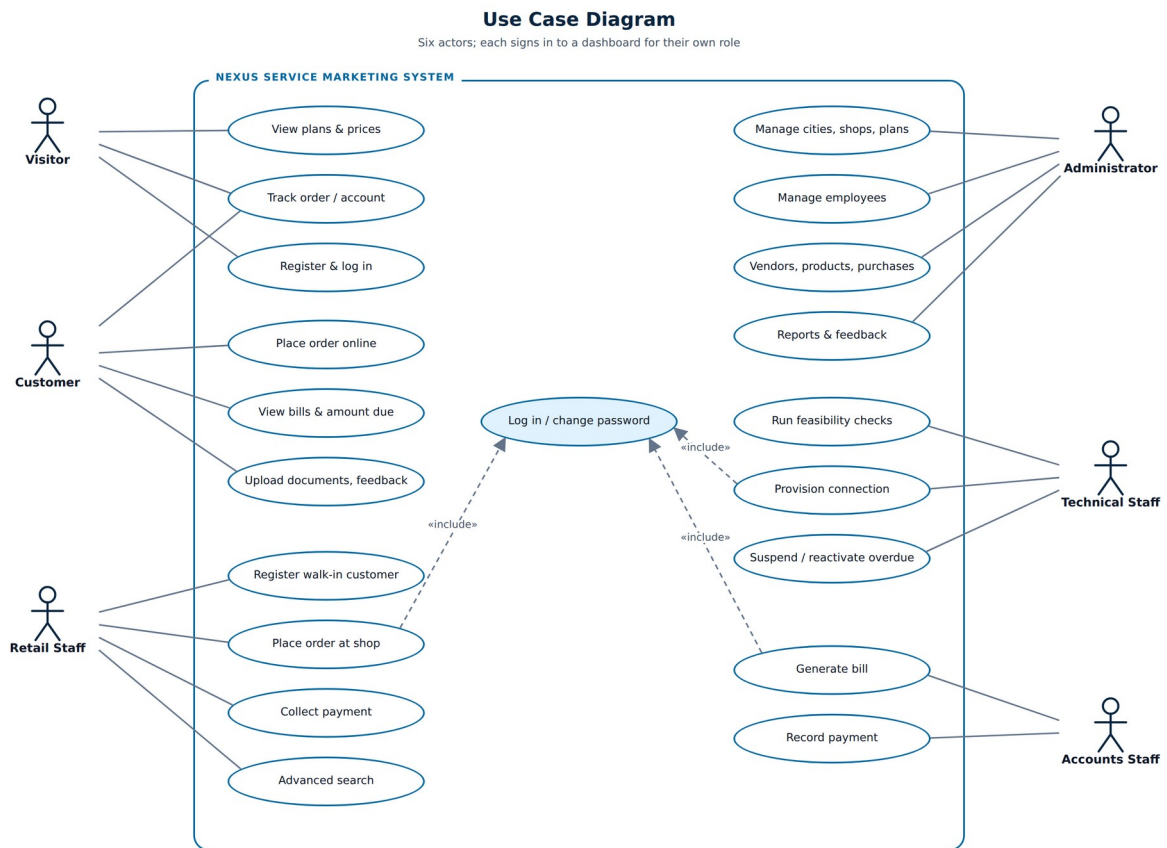


Figure 5. Use case diagram

6.4 Workflow flowchart

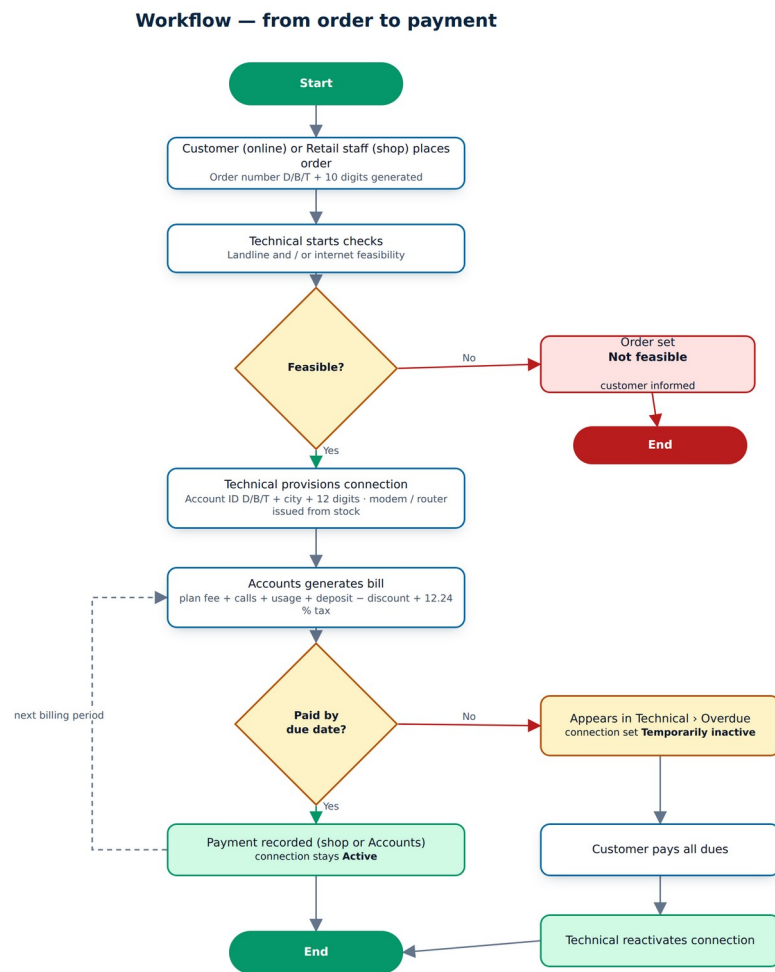


Figure 6. Order-to-payment workflow

6.5 Status life cycles

Table 12. Status life cycles

Object	Statuses (in order)
Order	Placed -> Under feasibility check -> Feasible / Not feasible -> Connected (or Cancelled before connection)
Feasibility check	Pending -> Feasible / Not feasible
Connection	Active <-> Temporarily inactive -> Permanently inactive
Bill	Issued -> Partially paid -> Paid (or Cancelled)

6.6 Algorithms

Order placement

1. Retail employee or customer selects connection type, plan, city, address and quantity.

2. The system checks that the plan matches the type and the city is active.
3. Connections already provided, orders in progress and the new order are counted; if the total is 10 or more the matching discount scheme is attached.
4. The next serial for the type is taken from IdentifierCounters and the order number is built (letter + 10 digits).
5. The order is saved with status Placed.

Feasibility check

1. Technical opens a Placed order and starts the checks.
2. Dial-Up: if the customer has an active Nexus landline only an internet check is created, otherwise landline and internet checks. Broadband: internet check. Telephone: landline check.
3. For each check the distance and, for internet, the server availability are recorded.
4. All checks feasible: the order becomes Feasible; any check not feasible: the order becomes Not feasible.

Connection creation

1. The order must be Feasible; equipment is chosen from stock and the stock quantity is reduced.
2. Security deposit = plan deposit, reduced by the discount percent when the scheme applies.
3. Account ID = type letter + city code + 12-digit serial. The connection starts Active and the order becomes Connected.

Bill generation

1. Accounts selects the connection and period; the system previews the plan fee, rates, unbilled replacements and brought-forward balance.
2. Call charges = minutes x per-minute rates; other usage is added as an amount.
3. First bill: add the security deposit and subtract the discount.
4. SubTotal = plan fee + calls + usage + deposit + replacements - discount; tax = 12.24 % of SubTotal; Total = SubTotal + tax.
5. Payments reduce the outstanding amount; the status moves Issued -> Partially paid -> Paid.

7. User Interface Design

The interface was designed as two experiences: a **marketing website** for visitors (immersive hero, 3D globe, plans) and a **work application** for signed-in users (sidebar, dashboards, dense tables). Both share one design system.

7.1 Design system

Table 13. Design system

Element	Choice
Colours	Primary blue #0369A1, navy #0A2540 (sidebar, hero), emerald #059669 (success / call to action), soft ice background #F3F8FC
Typography	Space Grotesk for headings and numbers, DM Sans for text
Status colours	Grey = placed / pending, Blue = in progress / issued, Amber = feasible / partially paid / temporarily inactive, Green = active / connected / paid, Red = overdue / not feasible / closed
Layout	Fixed sidebar (268 px) with sections per role, sticky glass top bar with breadcrumb, quick Track box and account menu; content up to 1480 px
Components	KPI cards with icon and count-up number, attention feed, bar chart, quick actions, badges, toasts, confirm overlay
Responsive	Below 992 px the sidebar becomes an off-canvas drawer opened by the menu button; cards stack on phones

7.2 Motion and overlays

- **Cut-overlay page transition** - three angled panels (emerald, blue, navy) sweep across the screen when a link is followed and sweep away on the next page, with the Nexus logo in the middle.
- **Confirm overlay** - Delete, Deactivate, Suspend, Replace and Cancel order open a blurred overlay with a warning icon instead of the browser's plain confirm box.
- **Toasts** - success messages slide in at the top right, count down and disappear; hovering pauses them.
- **Micro-interactions** - count-up dashboard numbers, growing chart bars, staggered reveal of cards and table rows, ripple on buttons, 3D tilt on tiles, loading spinner on submit to prevent double posts.
- All motion uses only transform and opacity and is switched off for users who prefer reduced motion.



Figure 7. Cut-overlay transition while moving between pages

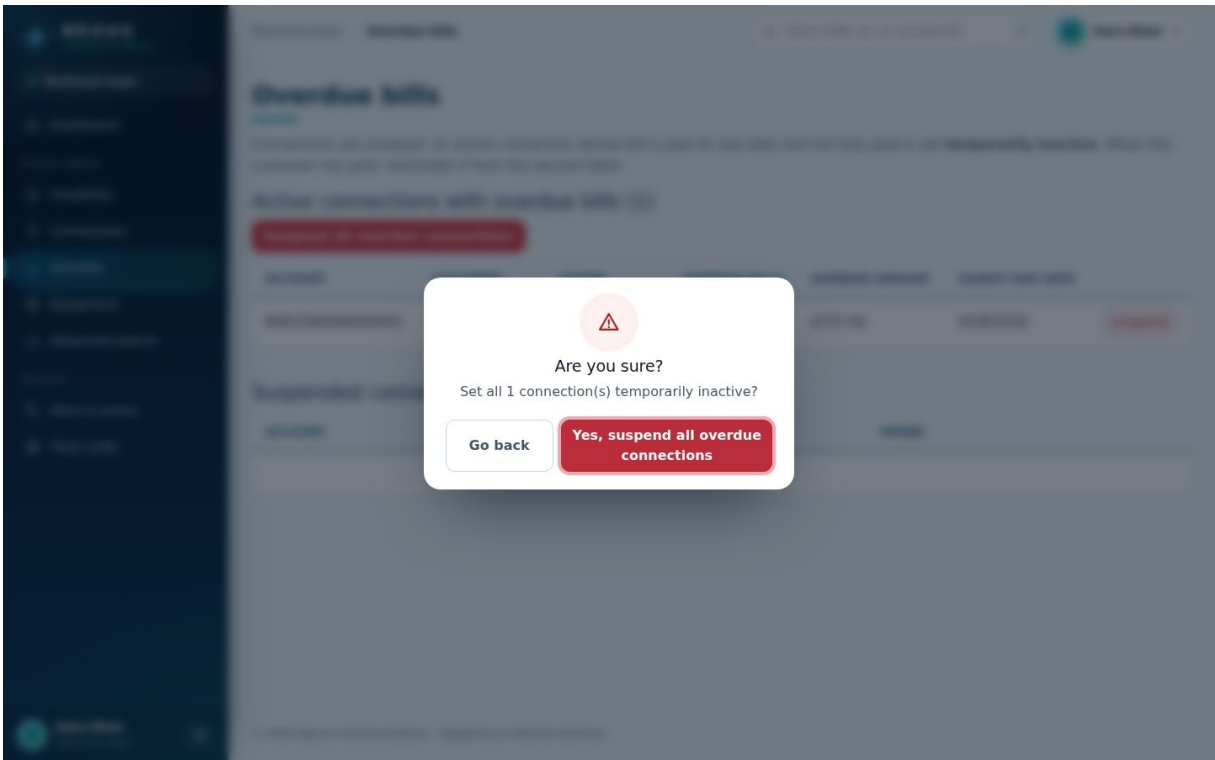


Figure 8. Confirm overlay before suspending overdue connections

Administrator > Cities

Q Track order no. or account ID / System Administrator

Cities

City 'Sukkur 473' created.

NAME	CODE	STATUS	
Faisalabad	005	Active	Edit
Islamabad	003	Active	Edit
Karachi	001	Active	Edit
Lahore	002	Active	Edit
Rawalpindi	004	Active	Edit
Sukkur 473	473	Active	Edit

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Figure 9. Success toast after saving a city

7.3 Public website

NEXUS Communications Home Plans Track Login Register

NOW CONNECTING EVERY CITY IN THE TERRITORY

Telephone & Internet, delivered close to home.

Dial-Up, Broadband and Landline connections with simple postpaid billing, a retail shop in every city and a self-service portal to follow your order from application to activation.

[View plans & prices](#) [Create customer account](#)

3 services: Dial-Up, Broadband, Landline

100% max bulk discount

24/7 online order & account tracking

Dial-Up

Hourly and unlimited 28/56 Kbps plans. Needs a Nexus landline - apply for both together.

Broadband

Hourly and unlimited 64/128 Kbps plans with a modem or router supplied.

Landline

Local and STD plans with clear per-minute call charges.

Figure 10. Home page

NEXUS Communications

Home

Plans

Track

Login

Register

Plans & prices

All prices are in US dollars. A refundable security deposit is charged on the first bill and a 12.24% service tax applies to the whole bill.

Dial-Up connection

A Nexus landline is required for a dial-up connection; it can be applied for together with the dial-up.

PLAN	DETAILS	VALIDITY	PRICE	SECURITY DEPOSIT
Dial-Up Hourly 10 Hrs	10 hours included	1 month	\$50.00	\$325.00
Dial-Up Hourly 30 Hrs	30 hours included	3 months	\$130.00	\$325.00
Dial-Up Hourly 60 Hrs	60 hours included	6 months	\$260.00	\$325.00
Dial-Up Unlimited 28 Kbps Monthly	Unlimited at 28 Kbps	1 month	\$75.00	\$325.00
Dial-Up Unlimited 56 Kbps Monthly	Unlimited at 56 Kbps	1 month	\$100.00	\$325.00
Dial-Up Unlimited 28 Kbps Quarterly	Unlimited at 28 Kbps	3 months	\$150.00	\$325.00
Dial-Up Unlimited 56 Kbps Quarterly	Unlimited at 56 Kbps	3 months	\$180.00	\$325.00

Broadband connection

A modem or router is supplied when the connection is activated.

PLAN	DETAILS	VALIDITY	PRICE	SECURITY DEPOSIT
Broadband Hourly 30 Hrs	30 hours included	1 month	\$175.00	\$500.00
Broadband Hourly 60 Hrs	60 hours included	6 months	\$315.00	\$500.00

Figure 11. Plans and prices

NEXUS Communications

Home

Plans

Track

Login

Register

ONE LOGIN FOR EVERY ROLE

Welcome back to

Nexus.

Admin, accounts, technical and retail staff, and customers -

sign in to reach your own dashboard.

Sign in

Username

Password

☐

Keep me signed in

Log in

Create a customer account

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Plans · Track · Privacy

Figure 12. Login (one login for every role)

Page 34 of 58

NEXUS Communications

HomePlansTrack

LoginRegister

Create Customer Account

Register to place orders and track your Nexus services.

Full name

Customer type

Individual

Email

Phone

Address

Postal code

CityId

Figure 13. Customer registration

NEXUS Communications

HomePlansTrack

LoginRegister

Track order / account

Order number (11 characters)

Account ID (16 characters)

Search

Order B0000000003

Broadband - Broadband Hourly 30 Hrs - Islamabad

Placed on08 Oct 2026

Last update08 Oct 2026

Connections requested1

Feasibility

Internet check: Feasible

Connections created

B003000000000001 - Active

Connected

Figure 14. Public order tracking

7.4 Administrator

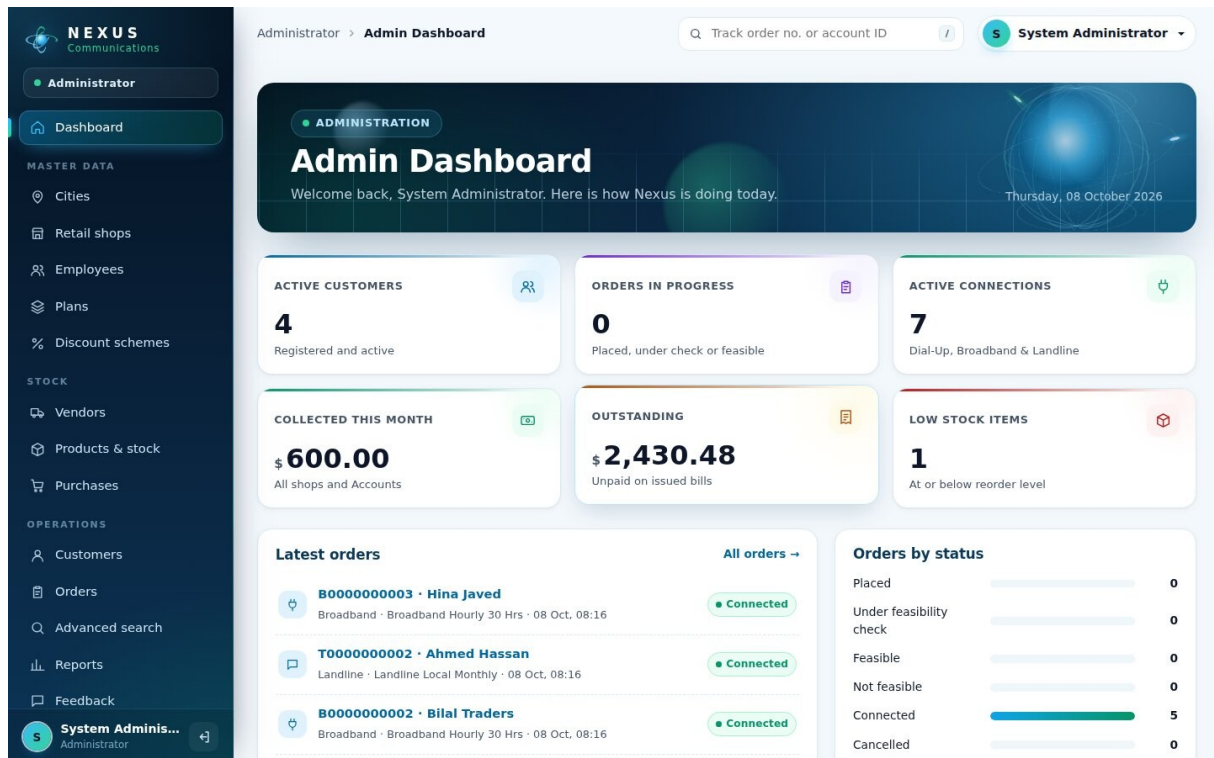


Figure 15. Admin dashboard with live figures

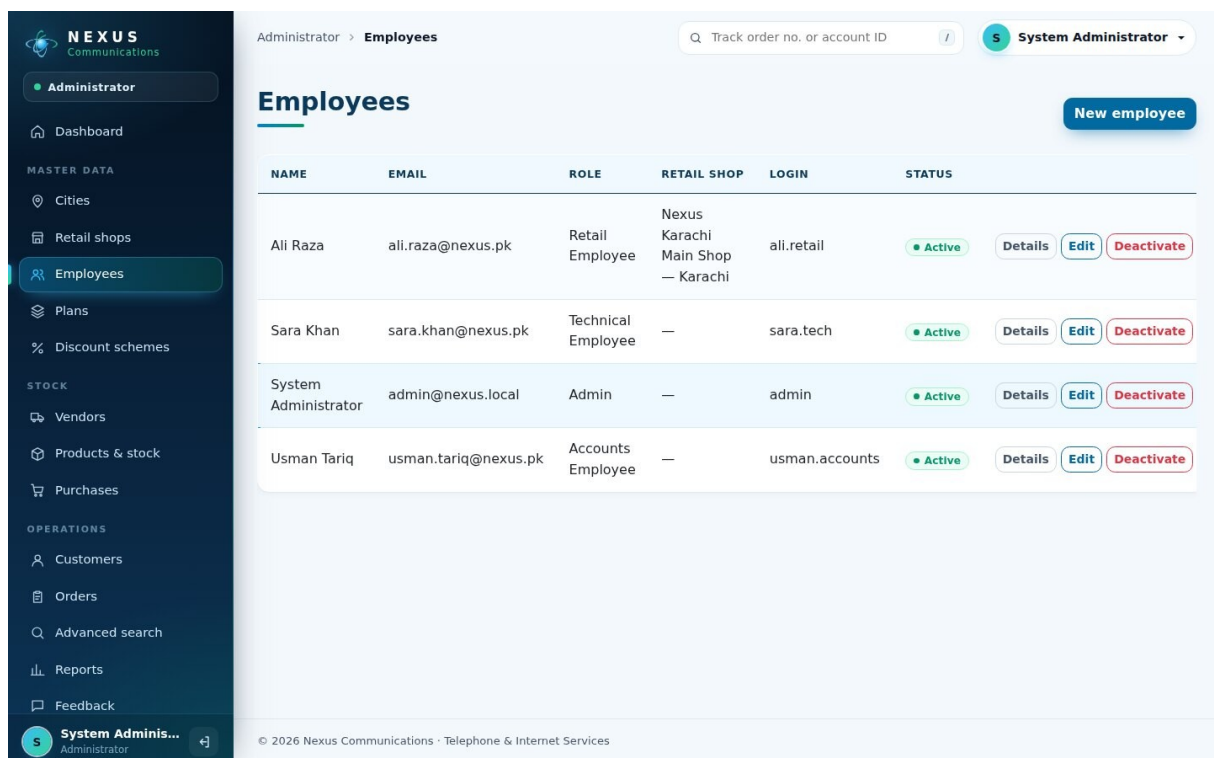


Figure 16. Employees and their logins



Administrator

Dashboard

MASTER DATA

Cities

Retail shops

Employees

Plans

Discount schemes

STOCK

Vendors

Products & stock

Purchases

OPERATIONS

Customers

Orders

Advanced search

Reports

Feedback

System Administrator

Administrator > New Employee

Q Track order no. or account ID / System Administrator

New Employee

This creates the employee record and login. The password is stored only as a salted hash.

Full name

Email

Phone

Role

Select a role

Login account

Username

Initial password

At least 8 characters. Share this initial password securely with the employee.

☒ IsActive

Figure 17. New employee form



Administrator

Dashboard

MASTER DATA

Cities

Retail shops

Employees

Plans

Discount schemes

STOCK

Vendors

Products & stock

Purchases

OPERATIONS

Customers

Orders

Advanced search

Reports

Feedback

System Administrator

Administrator > Plans

Q Track order no. or account ID / System Administrator

Plans

Search Clear

+ New Plan

NAME	TYPE	KIND	VALIDITY	PRICE	DEPOSIT	STATUS	
Broadband Hourly 30 Hrs	Broadband	Hourly	1 mo	\$175.00	\$500.00	Active	Edit Delete
Broadband Hourly 60 Hrs	Broadband	Hourly	6 mo	\$315.00	\$500.00	Active	Edit Delete
Broadband Unlimited 128 Kbps Monthly	Broadband	Unlimited	1 mo	\$350.00	\$500.00	Active	Edit Delete
Broadband Unlimited 128 Kbps Quarterly	Broadband	Unlimited	3 mo	\$445.00	\$500.00	Active	Edit Delete
Broadband Unlimited 64 Kbps Monthly	Broadband	Unlimited	1 mo	\$225.00	\$500.00	Active	Edit Delete
Broadband Unlimited 64 Kbps Quarterly	Broadband	Unlimited	3 mo	\$400.00	\$500.00	Active	Edit Delete
Dial-Up Hourly 10 Hrs	DialUp	Hourly	1 mo	\$50.00	\$325.00	Active	Edit Delete
Dial-Up Hourly 30 Hrs	DialUp	Hourly	3 mo	\$130.00	\$325.00	Active	Edit Delete
Dial-Up Hourly 60 Hrs	DialUp	Hourly	6 mo	\$260.00	\$325.00	Active	Edit Delete
Dial-Up Unlimited 28 Kbps	DialUp	Unlimited	1 mo	\$75.00	\$325.00	Active	Edit Delete

Figure 18. Plans with search and delete

Administrator > Products

Q Track order no. or account ID / System Administrator

Products

+ New Product

SKU	NAME	CATEGORY	VENDOR	STOCK	STATUS
RTR-ADSL	ADSL Wi-Fi Router	Router	TechLink Pvt Ltd	23	Active
MDM-56K	Dial-Up Modem 56K	Modem	TechLink Pvt Ltd	25	Active
TEL-HS1	Telephone Handset	Other	TechLink Pvt Ltd	3 (reorder!)	Active

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Figure 19. Products and stock (low-stock badge)

Administrator > All orders

Q Track order no. or account ID / System Administrator

All orders

Order, customer, email or p All statuses mm/dd/yyyy mm/dd/yyyy Search

ORDER	CUSTOMER	TYPE	PLAN	CITY	DATE	STATUS
B0000000003	Hina Javed	Broadband	Broadband Hourly 30 Hrs	Islamabad	10/8/2026	Connected
T0000000002	Ahmed Hassan	Telephone	Landline Local Monthly	Karachi	10/8/2026	Connected
B0000000002	Bilal Traders	Broadband	Broadband Hourly 30 Hrs	Karachi	10/8/2026	Connected
T0000000001	Fatima Noor	Telephone	Landline Local Monthly	Karachi	10/8/2026	Connected
B0000000001	Ahmed Hassan	Broadband	Broadband Hourly 30 Hrs	Karachi	10/8/2026	Connected

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Figure 20. All orders with filters

NEXUS
Communications

- Administrator
- Dashboard
- MASTER DATA
- Cities
- Retail shops
- Employees
- Plans
- Discount schemes
- STOCK
- Vendors
- Products & stock
- Purchases
- OPERATIONS
- Customers
- Orders
- Advanced search
- Reports
- Feedback
- System Admins...
Administrator

Administrator > Advanced search

S System Administrator

Advanced search

Find orders by order number and connections by account ID, or search by customer name, contact number or e-mail, service type, status and application date range.

Order number e.g. D000000001	Account ID 16-character account ID	Customer name, phone, or email Name, contact number or e-mail
Service type All services	Order status All order states	Connection status All connection states
Applied from mm/dd/yyyy	Applied through mm/dd/yyyy	Search Reset

Orders (5)

ORDER	CUSTOMER	CONTACT	TYPE	PLAN	APPLIED	STATUS
B0000000003	Hina Javed	03019998877	Broadband	Broadband Hourly 30 Hrs	08 Oct 2026	Connected
T0000000002	Ahmed Hassan	03331112233	Telephone	Landline Local Monthly	08 Oct 2026	Connected
B0000000002	Bilal Traders	02135556677	Broadband	Broadband Hourly 30 Hrs	08 Oct 2026	Connected
T0000000001	Fatima Noor	03452223344	Telephone	Landline Local Monthly	08 Oct 2026	Connected
B0000000001	Ahmed Hassan	03331112233	Broadband	Broadband Hourly 30 Hrs	08 Oct 2026	Connected

Figure 21. *Advanced search*


NEXUS
 Communications

Administrator

Dashboard

MASTER DATA

Cities

Retail shops

Employees

Plans

Discount schemes

STOCK

Vendors

Products & stock

Purchases

OPERATIONS

Customers

Orders

Advanced search

Reports

Feedback

System Administrator

Administrator > System reports

Track order no. or account ID

System Administrator

System reports

People and catalogue

Customers: 4 (4 active) · Employees: 4 · Shops: 5 · Plans: 18 · Products: 3 (1 low stock) · Vendors: 1

Orders

Total 5 · Pending 0 · Feasible 0 · Not feasible 0 · Connected 5

Connections

Total 7 · Active 7 · Temporarily inactive 0 · Permanently inactive 0

Billing

Bills 4 · Billed \$3,030.48 · Paid \$600.00 · Outstanding \$2,430.48 · Overdue bills 1 · Payments 1

Equipment demand (modems and routers)

Needed by open internet orders: 0 · In stock: 48 · Stock covers all open orders

Feedback

Responses 1 · Average rating 5.0 / 5

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Figure 22. *System reports*

7.5 Retail staff

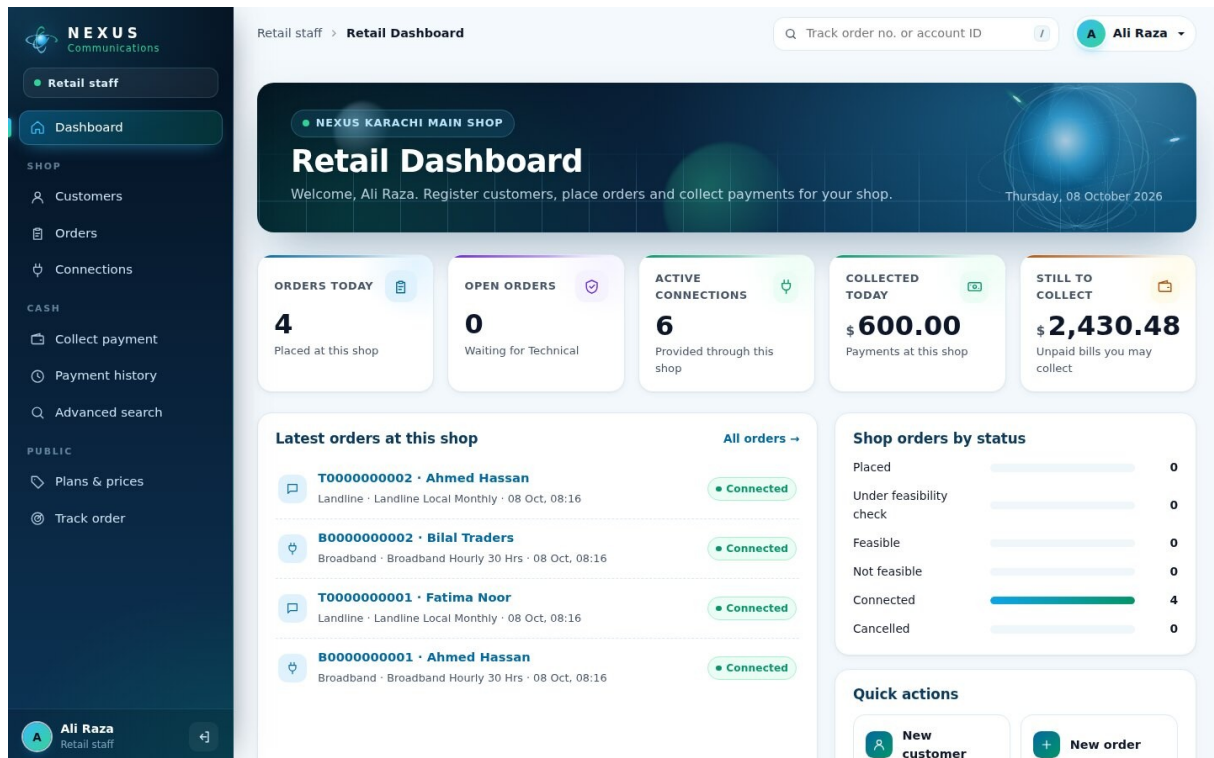


Figure 23. Retail dashboard for one shop

The screenshot shows the 'New Customer' registration form. The sidebar is identical to the previous figure. The main content area has a heading 'New Customer' and a subheading 'Create the customer profile and login, then continue to order placement.' The form fields include: Full name (text input), Customer type (dropdown menu with 'Individual' selected), Email (text input), Phone (text input), Address (text input), Postal code (text input), and CityId (dropdown menu with 'Select a city' selected). The user 'Ali Raza' is logged in, as indicated by the profile icon and name in the top right corner.

Figure 24. Register a walk-in customer

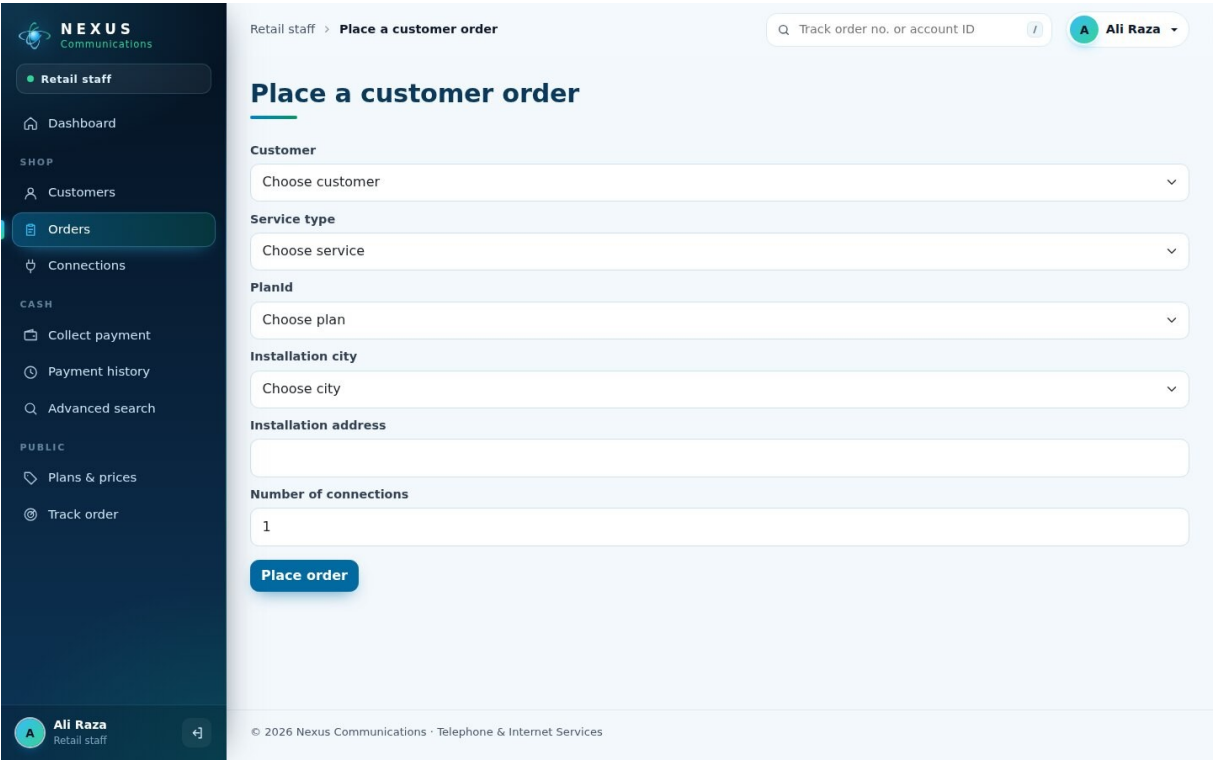


Figure 25. Place an order

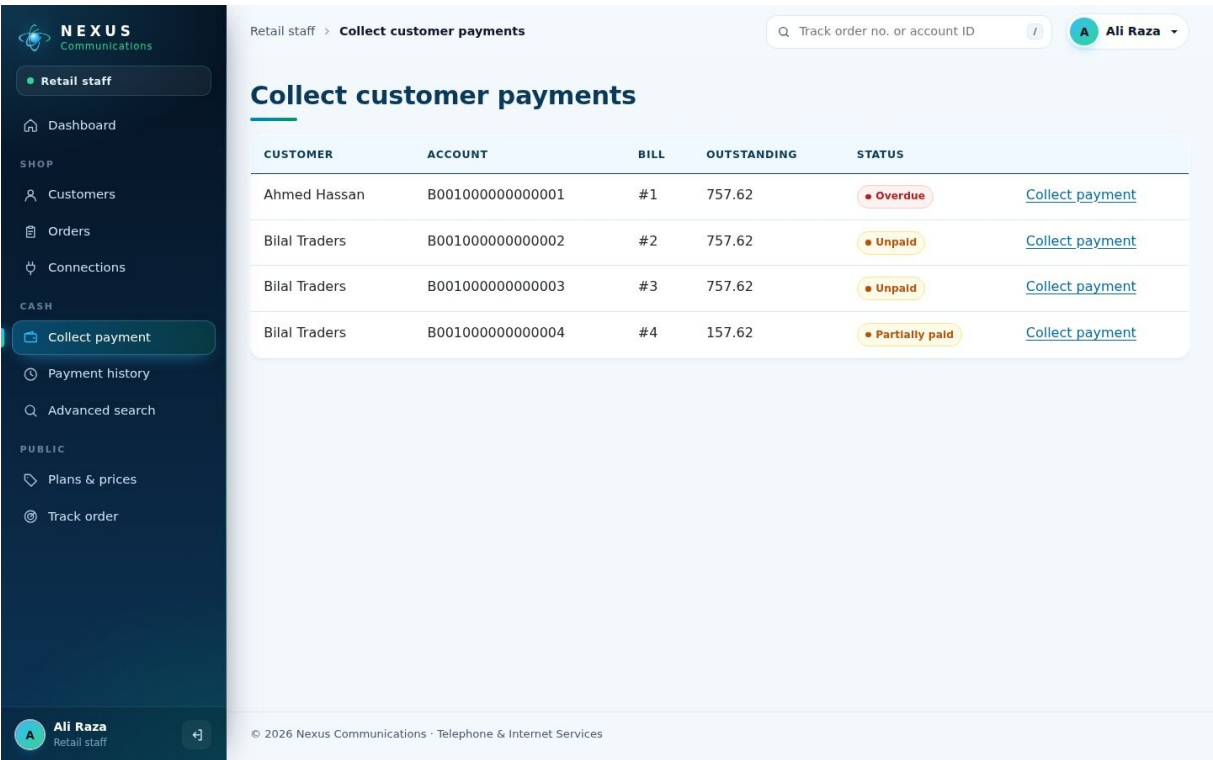


Figure 26. Collect customer payments

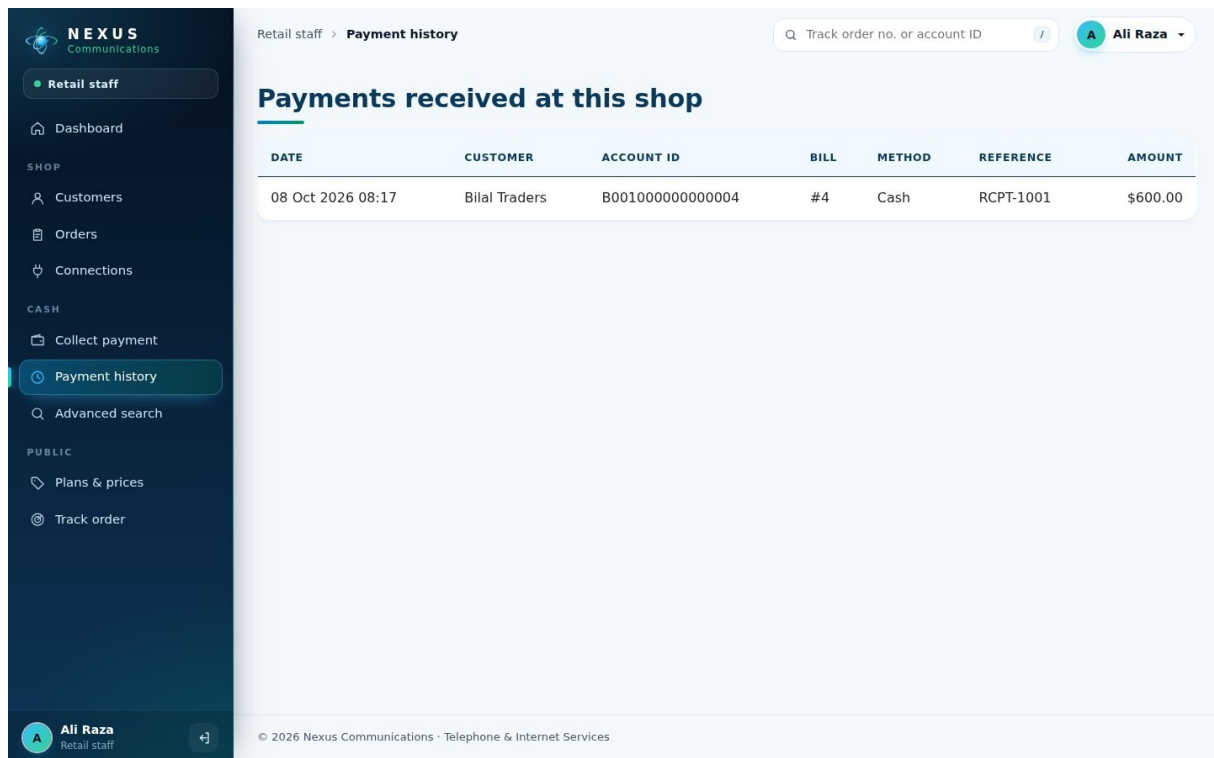


Figure 27. Payment history

7.6 Technical team

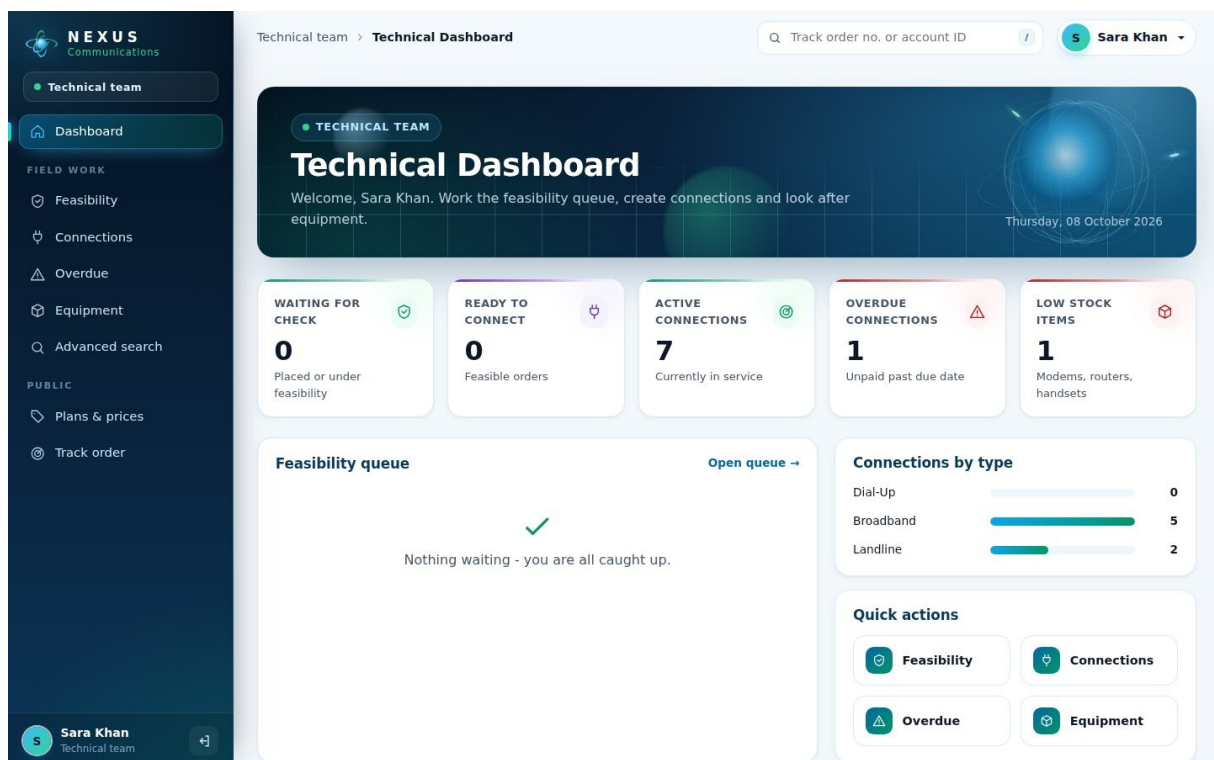


Figure 28. Technical dashboard

Technical team

Dashboard

FIELD WORK

Feasibility

Connections

Overdue

Equipment

Advanced search

PUBLIC

Plans & prices

Track order

Sara Khan

Technical team

Technical team > Orders and feasibility

Track order no. or account ID

Sara Khan

Orders and feasibility

ORDER	CUSTOMER	SERVICE	STATUS	CHECKS	ACTIONS
B0000000003	Hina Javed	Broadband	Connected	Internet: Feasible	
T0000000002	Ahmed Hassan	Telephone	Connected	Landline: Feasible	
B0000000002	Bilal Traders	Broadband	Connected	Internet: Feasible	
T0000000001	Fatima Noor	Telephone	Connected	Landline: Feasible	
B0000000001	Ahmed Hassan	Broadband	Connected	Internet: Feasible	

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Figure 29. Orders and feasibility

Technical team

Dashboard

FIELD WORK

Feasibility

Connections

Overdue

Equipment

Advanced search

PUBLIC

Plans & prices

Track order

Sara Khan

Technical team

Technical team > Connections

Track order no. or account ID

Sara Khan

Connections

1 active connection(s) have overdue bills. [Review and suspend](#)

Set a connection temporarily or permanently inactive, or move it to another plan of the same type. The security deposit is refundable when a connection is permanently closed.

ACCOUNT	CUSTOMER	TYPE	PLAN	STATUS	DEPOSIT	ORDER	STATUS UPDATE	CHANGE
B003000000000001	Hina Javed	Broadband	Broadband Hourly 30 Hrs	Active	\$500.00	B0000000003	Active <button>Sav</button>	Broa
T001000000000002	Ahmed Hassan	Telephone	Landline Local Monthly	Active	\$250.00	T0000000002	Active <button>Sav</button>	Land
B001000000000004	Bilal Traders	Broadband	Broadband Hourly 30 Hrs	Active	\$500.00	B0000000002	Active <button>Sav</button>	Broa
B001000000000003	Bilal Traders	Broadband	Broadband Hourly 30 Hrs	Active	\$500.00	B0000000002	Active <button>Sav</button>	Broa
B001000000000002	Bilal Traders	Broadband	Broadband Hourly 30 Hrs	Active	\$500.00	B0000000002	Active <button>Sav</button>	Broa
T001000000000001	Fatima	Telephone	Landline Local	Active	\$250.00	T0000000001	Active <button>Sav</button>	Land

Figure 30. Connections: status, plan change, equipment replacement

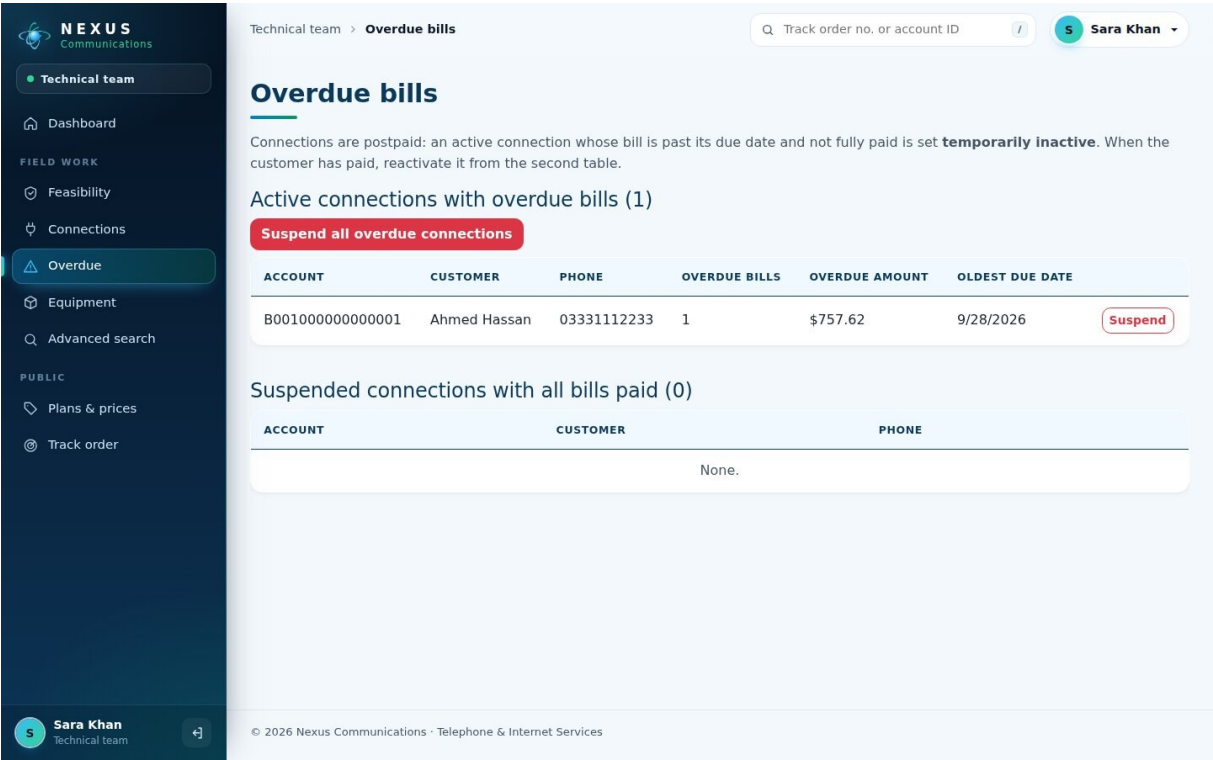


Figure 31. Overdue connections

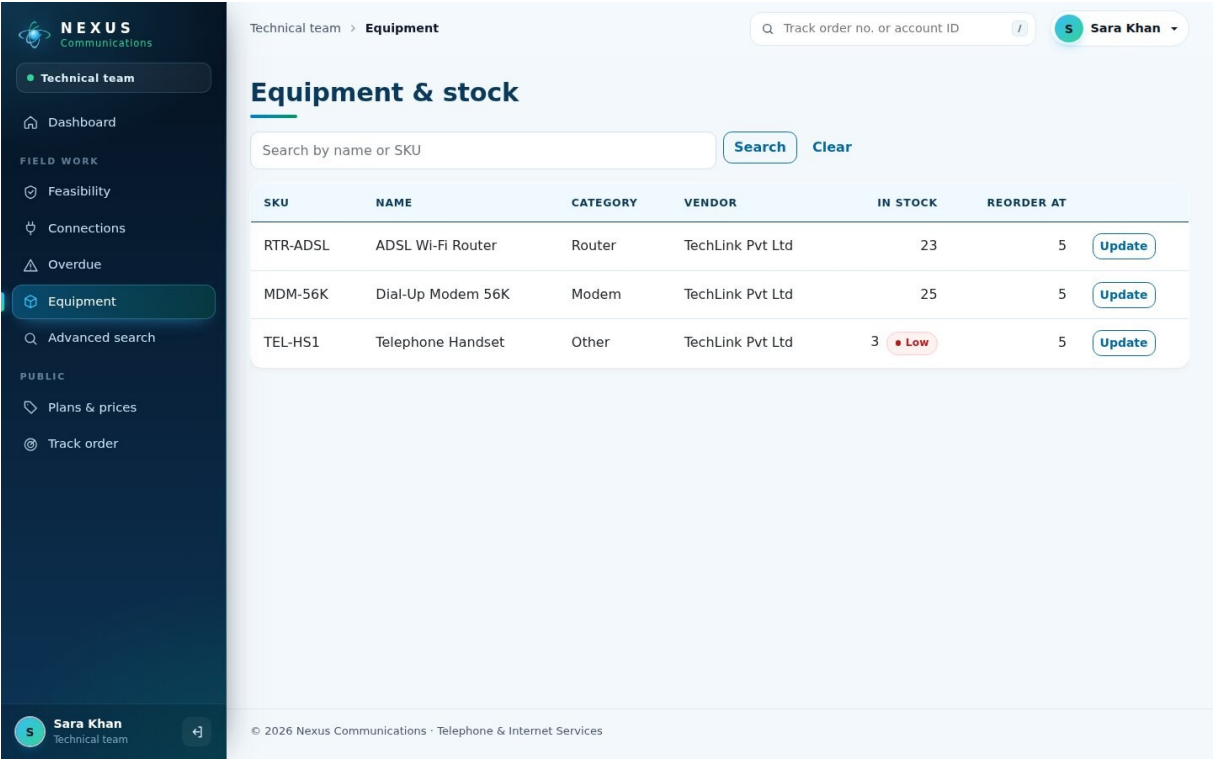


Figure 32. Equipment and stock

7.7 Accounts

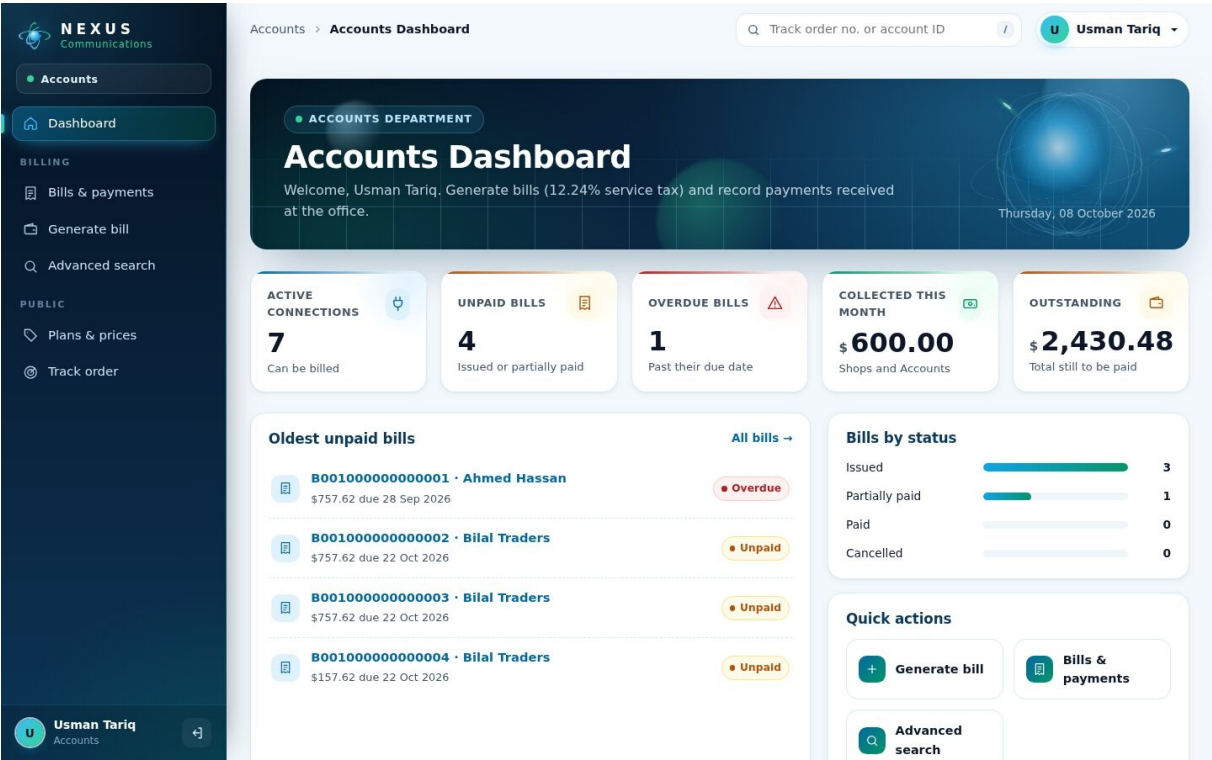


Figure 33. Accounts dashboard

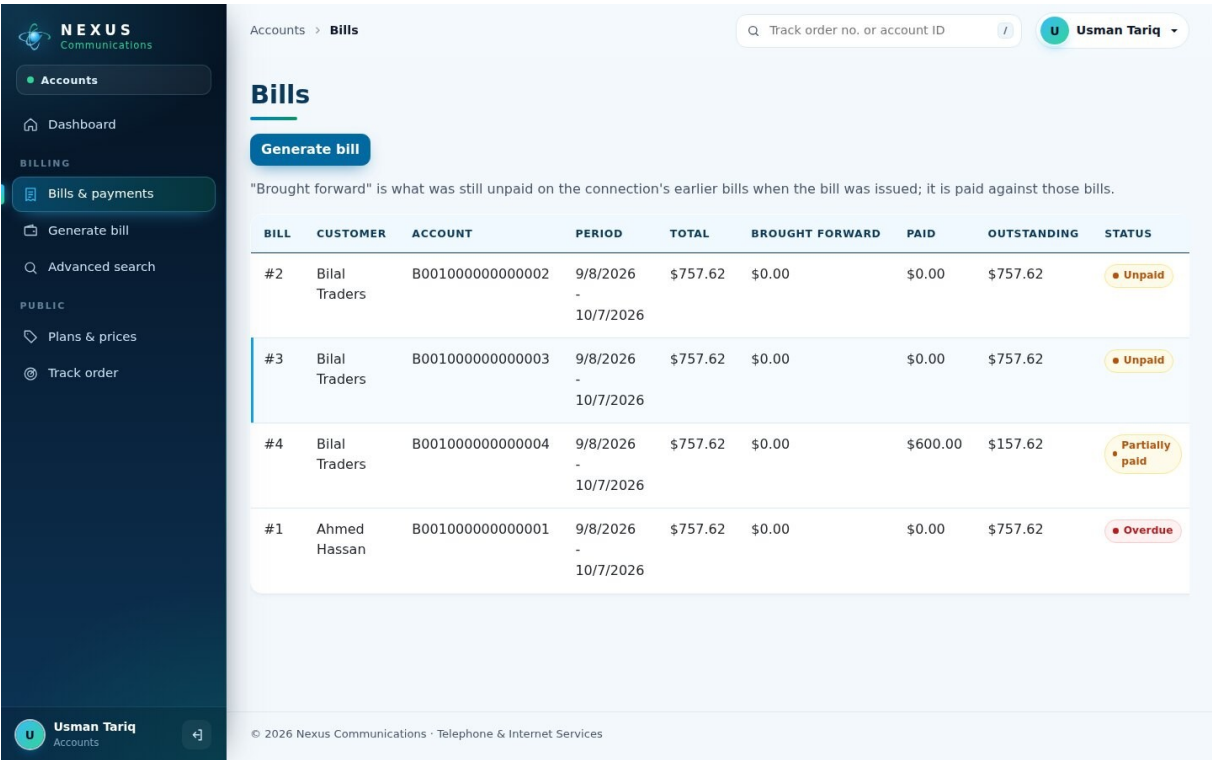
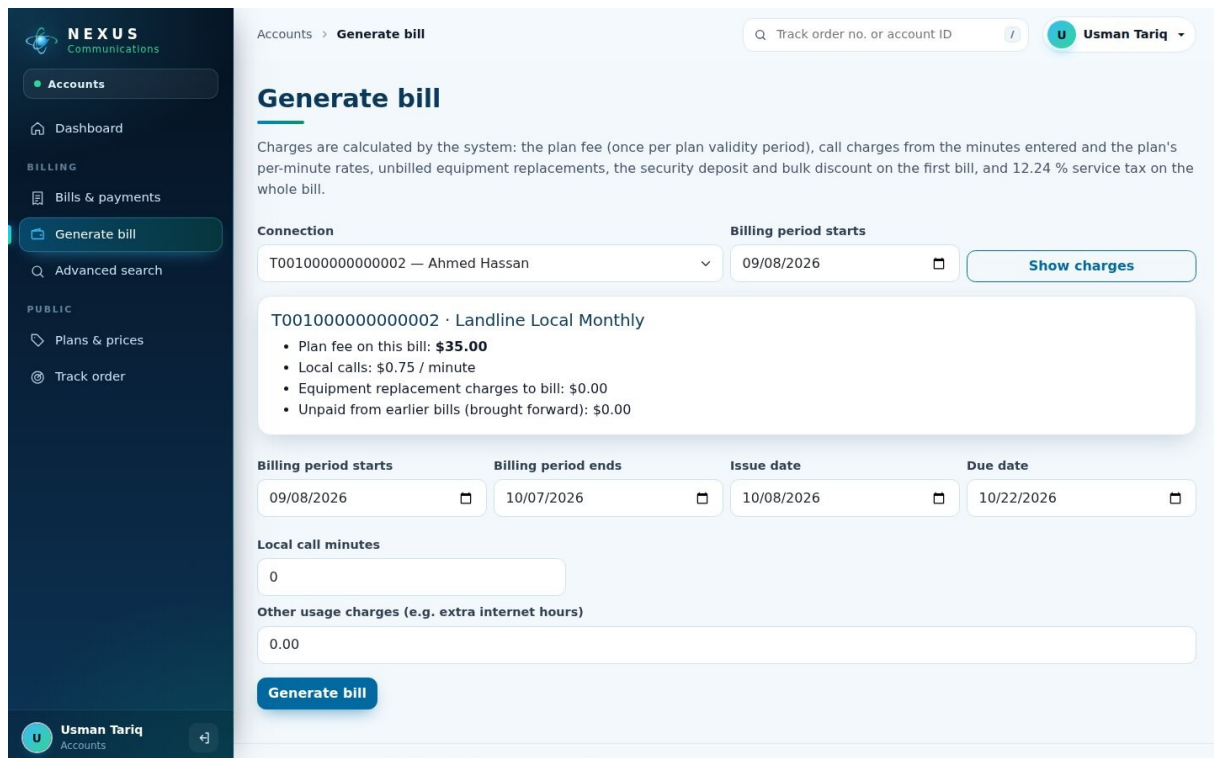


Figure 34. Bills and outstanding amounts



NEXUS Communications

Accounts > **Generate bill**

Search: Track order no. or account ID / **U Usman Tariq**

Generate bill

Charges are calculated by the system: the plan fee (once per plan validity period), call charges from the minutes entered and the plan's per-minute rates, unbilled equipment replacements, the security deposit and bulk discount on the first bill, and 12.24 % service tax on the whole bill.

Connection T001000000000002 — Ahmed Hassan **Billing period starts** 09/08/2026 **Show charges**

T001000000000002 · Landline Local Monthly

- Plan fee on this bill: **\$35.00**
- Local calls: \$0.75 / minute
- Equipment replacement charges to bill: \$0.00
- Unpaid from earlier bills (brought forward): \$0.00

Billing period starts 09/08/2026 **Billing period ends** 10/07/2026 **Issue date** 10/08/2026 **Due date** 10/22/2026

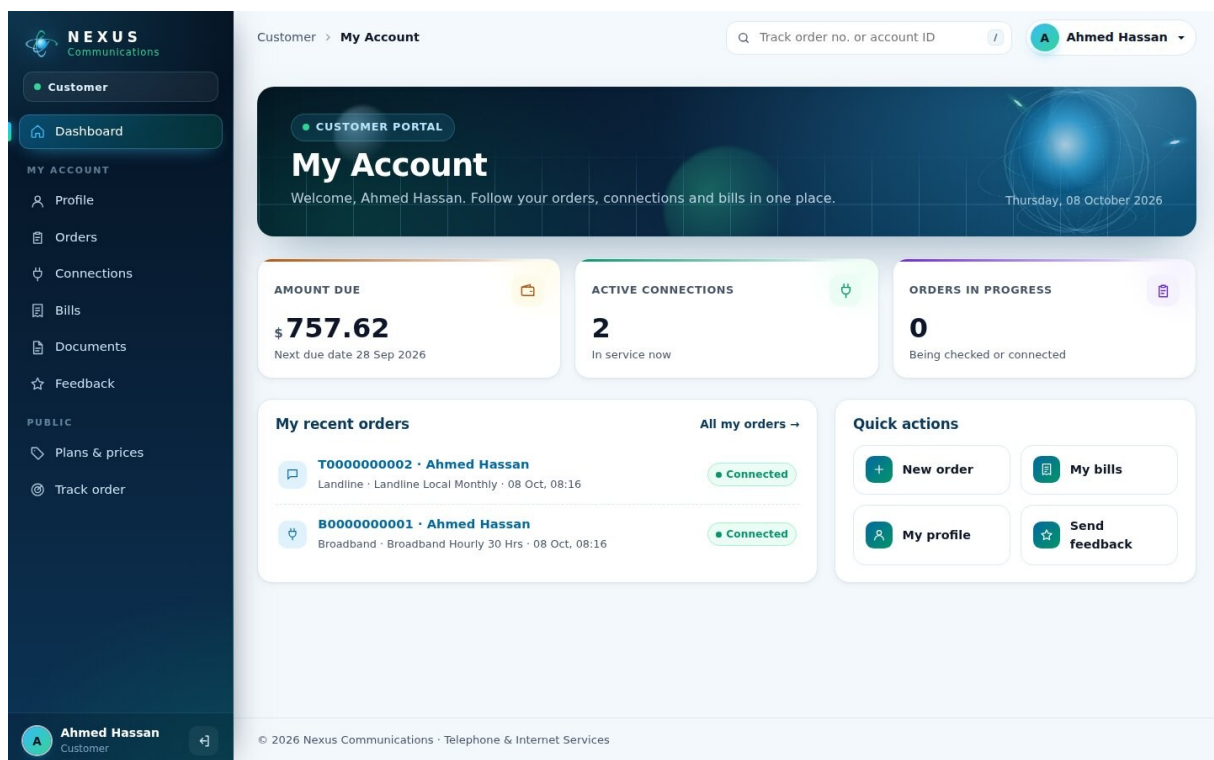
Local call minutes 0

Other usage charges (e.g. extra internet hours) 0.00

Generate bill

Figure 35. Generate a bill with charge preview

7.8 Customer portal



NEXUS Communications

Customer > **My Account**

Search: Track order no. or account ID / **A Ahmed Hassan**

My Account

Welcome, Ahmed Hassan. Follow your orders, connections and bills in one place. Thursday, 08 October 2026

AMOUNT DUE \$757.62 Next due date 28 Sep 2026

ACTIVE CONNECTIONS 2 In service now

ORDERS IN PROGRESS 0 Being checked or connected

My recent orders All my orders →

- T0000000002 · Ahmed Hassan Landline · Landline Local Monthly · 08 Oct, 08:16 Connected
- B0000000001 · Ahmed Hassan Broadband · Broadband Hourly 30 Hrs · 08 Oct, 08:16 Connected

Quick actions

- New order
- My bills
- My profile
- Send feedback

Figure 36. Customer dashboard with amount due

Customer > My service orders

Q Track order no. or account ID / H Hina Javed

My service orders

[Place an order](#)

ORDER	SERVICE	PLAN	PLACED	STATUS	
B0000000003	Broadband	Broadband Hourly 30 Hrs	10/8/2026	Connected	Track

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Figure 37. My orders

Customer > My connections

Q Track order no. or account ID / H Hina Javed

My connections

Your account IDs, connection status and the amount currently due.

ACCOUNT ID	SERVICE	PLAN	CITY	STATUS	AMOUNT DUE
B003000000000001	Broadband	Broadband Hourly 30 Hrs	Islamabad	Active	\$0.00

[Place a new order](#)

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Figure 38. My connections

NEXUS Communications

Customer > **My profile**

Q Track order no. or account ID

H Hina Javed

My profile

FullName
Hina Javed

Email
hina.javed@gmail.com

Phone
03019998877

AddressLine
Street 7, F-8/3

PostalCode
44000

CityName
Islamabad

Save

H Hina Javed
Customer

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Figure 39. My profile

7.9 Mobile view

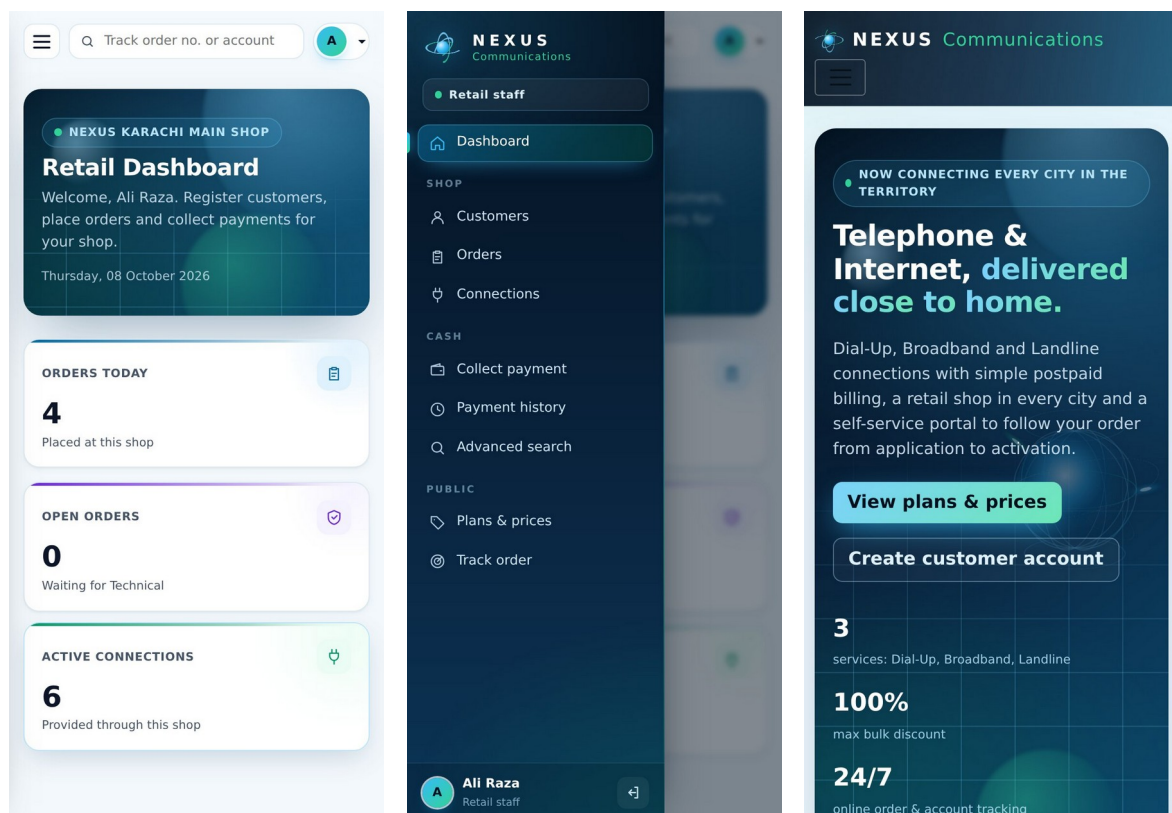


Figure 40. Phone: dashboard, sidebar drawer and home page

8. Testing

Testing combined manual checks with an automated browser run (Playwright, Chromium) that signs in as every role and performs the complete workflow through the real screens - the same screens shown in chapter 7. Negative tests check validation, security and the new overlays.

8.1 System test cases

Table 14. Test cases

ID	Test	Expected result	Result
TC-01	Admin creates vendor, 3 products, a purchase and 3 employees (retail, technical, accounts)	Records saved, purchase increases stock	Pass
TC-02	Retail staff registers 2 individual and 1 corporate customer	Customers and logins created	Pass
TC-03	Corporate customer without company name	Error "Company name is required for a corporate customer"	Pass
TC-04	Retail places Broadband and Telephone orders (one with quantity 3)	Order numbers B0000000001, T0000000001 ... generated	Pass
TC-05	Visitor registers, logs in and places an order online	Order B0000000003 for the customer	Pass
TC-06	Technical starts checks and records 5 feasibility results	Orders become Feasible	Pass
TC-07	Multi-connection order provisioned with one serial number	Error: enter serials individually after provisioning	Pass
TC-08	Technical provisions all feasible orders	7 connections with 16-character account IDs; stock reduced	Pass
TC-09	Accounts generates 4 bills (one back-dated)	Bills with plan fee, calls, deposit and 12.24 % tax; back-dated bill overdue	Pass
TC-10	Retail collects more than the outstanding amount	Error "Payment exceeds the outstanding balance"	Pass
TC-11	Retail collects a partial payment	Bill becomes Partially paid; history updated	Pass
TC-12	Customer sends feedback	Feedback listed for Admin with rating	Pass
TC-13	Login with a wrong password	"Invalid username or password."	Pass
TC-14	Retail staff opens /Admin/Cities	Redirected to Access denied	Pass
TC-15	Anonymous user opens /Accounts/Bills	Redirected to Login	Pass
TC-16	Track an order number that does not exist	"No order was found for B9999999999"	Pass
TC-17	Quick Track box with "ABC" / with "b0000000001"	Box shakes and stays / opens the order	Pass
TC-18	New city with an existing code 001	"A city with this code already exists."	Pass
TC-19	New city with code "12A"	"City code must be exactly 3 digits"	Pass
TC-20	Deactivate employee, then "Go back" in the confirm overlay	Nothing changes	Pass

ID	Test	Expected result	Result
TC-21	Employee with a 3-character password	Minimum length 8 error	Pass
TC-22	Registration with different password and confirmation	"Confirm password and Password do not match"	Pass
TC-23	Customer opens another customer's order by URL	HTTP 404 - not shown	Pass
TC-24	Technical suspends overdue connection, customer pays, Technical reactivates	Status Active -> Temporarily inactive -> Active	Pass

8.2 Unit testing checklist (development phase)

Checks recorded during development of the business services, kept here for completeness:

Table 15. Unit testing checklist

No.	Test	Result
1	Order ID is 11 characters and starts with D, T or B	Pass
2	Account ID is 16 characters with the correct city code	Pass
3	Dial-Up order for a customer without landline creates two feasibility checks	Pass
4	Dial-Up order with a Nexus landline creates one internet check	Pass
5	Quantity 10 to 14 attaches the 25 % discount	Pass
6	A connection cannot be created for a non-feasible order	Pass
7	Stock reduces when equipment is issued	Pass
8	Bill adds 12.24 % service tax	Pass
9	Retail employee can collect only bills their shop may collect	Pass
10	A permanently inactive connection cannot be reactivated	Pass
11	Advanced search filters by name, type, date and contact number	Pass
12	Customer can cancel an order before connection	Pass

Result

All 24 test cases passed on SQL Server 2022 with .NET 8.

System test cases TC-01 to TC-24 were run in this release; the schema was checked against the entity model: 19 tables, 5 migrations, 47 CHECK constraints.

9. Installation Guide

9.1 Requirements

- Windows 10 / 11 with **.NET 8 SDK** (`dotnet --version` shows 8.x).
- SQL Server** 2019 / 2022 (Express, Developer or Standard) and SQL Server Management Studio.
- Visual Studio 2022 (optional) and Google Chrome.

9.2 Connect SQL Server

In SSMS choose **Database Engine**, Server name `.` (this computer), **SQL Server Authentication**, Login `sa`, Password `aptech`, Encryption **Mandatory** with **Trust server certificate** ticked, then Connect.

The project already contains this connection in `appsettings.json`:

`appsettings.json`

```
"DefaultConnection": "Server=.;Database=NexusServiceMarketingDb;User
Id=sa;Password=aptech;MultipleActiveResultSets=true;TrustServerCertificate=True"
For SQL Express write Server=.\SQLEXPRESS (the backslash is doubled inside JSON).
```

9.3 Create the database

Open PowerShell in the folder that contains `NexusServiceMarketingSystem.csproj` and run the commands one at a time:

Table 16. *Commands*

Step	Command	What it does
1	<code>dotnet tool install --global dotnet-ef --version 8.*</code>	Installs the EF Core tool (first time only; use "update" if installed)
2	<code>dotnet ef dbcontext info</code>	Shows the server it will use - Data source must be "."
3	<code>dotnet ef database update</code>	Applies the 5 migrations: database, 19 tables, plans and discount bands
4	<code>dotnet run</code>	Starts the website at <code>http://localhost:5294</code>

Without dotnet-ef: in SSMS run `CREATE DATABASE NexusServiceMarketingDb;`, select the new database, open `Database/NexusServiceMarketingDb-schema.sql` and press F5. The website also applies migrations automatically when it starts.

9.4 First login

Table 17. *Default login*

Role	Username	Password
Administrator (created automatically)	admin	Admin@123

After the first login: change the admin password, check the 5 starter cities and shops, create employees (Retail, Technical, Accounts) and start taking orders.

9.5 Troubleshooting

Table 18. *Troubleshooting*

Message	Solution
Login failed for user 'sa'	Check the password; in SSMS > Server properties > Security choose "SQL Server and Windows Authentication mode" and restart the SQL Server service.
A network-related or instance-specific error	Wrong server name - use "." or ".\SQLEXPRESS".
Database created but not visible in SSMS	Run "dotnet ef dbcontext info"; the Data source must be the same server SSMS is connected to. Refresh the Databases folder.
dotnet-ef is not recognized	Close and reopen the terminal after installing, or use the SQL script method.
Port already in use	Close the other running copy, or run: dotnet run --urls http://localhost:5300

10. User Guide

10.1 Getting around

- After login every user lands on their **dashboard**. The sidebar lists only the pages of that role; the current page is highlighted.
- Type an order number or account ID in the **Track** box at the top and press Enter (press / to jump to the box).
- Click any order number or account ID in a table to copy it.
- On a phone, open the menu with the button at the top left.

10.2 Administrator

1. Master data > **Cities**: add each city with its unique 3-digit code.
2. **Retail shops**: add at least one shop per city.
3. **Employees**: create Retail (choose the shop), Technical and Accounts staff with their username and first password.
4. **Plans** and **Discount schemes**: check prices; plans already used cannot be deleted, deactivate them instead.
5. Stock > **Vendors, Products, Purchases**: keep modems and routers in stock (low-stock items show in red).
6. Operations > Orders, Advanced search, **Reports** and Feedback to monitor the business.

10.3 Retail staff

1. **Customers > New customer** for a walk-in customer (corporate customers need a company name).
2. **Orders > Place order**: choose customer, service, plan, city, address and number of connections. For dial-up without a Nexus line, also choose a landline plan.
3. **Collect payment**: open a bill, enter the amount (full or partial), method and reference.
4. **Connections** and **Payment history** show everything handled by the shop.

10.4 Technical team

1. **Feasibility**: press *Start checks* on a new order, then *Record result* for each check.
2. For feasible orders press *Provision*: choose the modem / router, serial and (for landlines) phone numbers.
3. **Connections**: change status, change plan or replace spoiled equipment.
4. **Overdue**: suspend connections with overdue bills; reactivate them when the customer has paid.

10.5 Accounts

1. **Generate bill**: choose the connection and period start, press *Show charges*, enter minutes and other usage, then *Generate bill*.
2. **Bills & payments**: see outstanding amounts and record payments received at the office.

10.6 Customer

1. **Register** on the website, then log in.
2. **Orders > New order** to apply for a connection and follow its status.
3. The dashboard shows the **amount due** and next due date; **Bills** lists every bill and payment.
4. Upload documents and send feedback from the sidebar.

11. Developer Guide

11.1 Folder structure

Table 19. Project structure

Folder / file	Contents
Areas/<Role>/Controllers, Views	Admin, Retail, Technical, Accounts, Customer areas
Controllers, Views	Public pages: Home, Catalog, Track, Account, Search
Services/	Workflows (orders, provisioning, billing, customers), Dashboard, Identifiers, Security, Documents
Data/	AppDbContext, design-time factory, converters, seeders
Migrations/	Five EF Core migrations and the model snapshot
Models/	Entities (19), enums, view models, dashboard models, NexusIcons
Views/Shared/	_Layout (app shell), _SideNav, Dashboard, partials
wwwroot/css	site.css (design tokens, public site, components), nexus-app.css (shell, dashboards, overlays)
wwwroot/js	site.js (loader, reveal, tilt, forms), nexus-app.js (sidebar, quick track, toasts, confirm, badges, counters)
Database/	Full SQL script and Run-Migration.bat

11.2 Adding a feature

1. Add or change the entity in `Models/Entities` and its configuration in `AppDbContext`.
2. Create a migration: `dotnet ef migrations add <Name>` then `dotnet ef database update`.
3. Put the rule in a service (never in the controller), register it in `Program.cs`.
4. Add the controller action with `[Authorize(Roles = ...)]` and `[ValidateAntiForgeryToken]` on POST.
5. Add the view; add a sidebar link in `Views/Shared/_SideNav.cshtml`.

12. Project Management

12.1 Task sheet

Table 20. *Task sheet*

No.	Task	Status
1	Requirement analysis	Completed
2	Database design and migrations	Completed
3	Admin module	Completed
4	Retail module	Completed
5	Technical module	Completed
6	Accounts module	Completed
7	Customer portal and public tracking	Completed
8	App shell, dashboards, motion and overlays	Completed
9	Testing	Completed
10	Documentation, presentation and submission	Completed

12.2 Project review and monitoring report

Table 21. *Review and monitoring*

Review	Progress	Issues and action
1st status	Database, migrations and Admin module ready	Database moved from LocalDB to SQL Server by changing the connection string
2nd status	All five role areas, billing, search, UI theme and documents ready	Empty city list fixed by starter data; plan change and order cancel added
Final	Sidebar app shell, live role dashboards, cut-overlay transitions, confirm overlays, documentation and presentation	Connection set to the local SQL Server (sa login); full workflow and negative tests passed

12.3 Final checklist

Table 22. *Final checklist*

Item	Done
Source code (ZIP)	Yes
Database scripts and migrations	Yes
Database backup (.bak) - SSMS > NexusServiceMarketingDb > Tasks > Back Up	Do before submitting
Login credentials documented	Yes
Project documentation (this document, Word and PDF)	Yes
Status report and feedback form (fill in your details)	Yes
Animated presentation	Yes

13. Conclusion and Future Enhancements

The Nexus Service Marketing System meets the problem statement: every order has a unique number and moves through feasibility, connection, billing and payment in one shared SQL Server database; every role has its own secure dashboard; and bills are calculated by the system with the required discount and tax rules. The new interface - sidebar shell, live dashboards, cut-overlay transitions and confirm overlays - makes the system modern and quick to use.

Future enhancements

- Online payments through a payment gateway and automatic receipts by e-mail / SMS.
- Call detail records imported from the exchange so bills calculate minutes automatically.
- Stock movement ledger and connection status history for full audit trails.
- Charts of monthly revenue and new connections per city on the admin dashboard.
- A mobile app for field technicians with photo upload at installation.

Demo logins used in this document

Admin: `admin` / `Admin@123`

Retail: `ali.retail` / `Nexus@12345` - Technical: `sara.tech` / `Nexus@12345` - Accounts: `usman.accounts` / `Nexus@12345`

Customers: `ahmed.hassan`, `hina.javed` / `Customer@123` (created during testing; a fresh database contains only the admin)